



Internship Report

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Institution – St Joseph's University

Organization Name – Lohia Corp Limited

TITLE - Commissioning Panel Of BB1500

Duration – 14 May 2025 to 20 June 2025

Department – Electronics and Computer Science

Submitted To – Adivera Pandian

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Acknowledgement

First and foremost, I am thankful to **St Joseph's University** and the Department of Electrical for giving me the opportunity to pursue this internship as part of my academic curriculum. I am especially grateful to **Mrs Regina Methias (Pro Vice Chancellor)**, whose encouragement and constant guidance enabled me to begin and complete this internship successfully.

I extend my sincere thanks to the management and entire team at **LohiaCorp** for welcoming me as an intern and for giving me the chance to explore the practical aspects of Electrical and machine circuit systems. I am truly thankful to my mentor at the company, **Mr Adiveera Pandian, Director of Manufacturing** for his patient explanations, real-time demonstrations, and continuous support throughout the internship period.

This internship has not only enhanced my technical knowledge but also improved my confidence in handling real-world applications. I appreciate the valuable insights and the working environment provided by the organization, which allowed me to learn beyond textbooks.



Introduction

Objective Of the Report -

To Get Exposure On electrical circuit diagram produced from EPLAN software & Field Wiring, Panel Wiring and Troubleshooting.

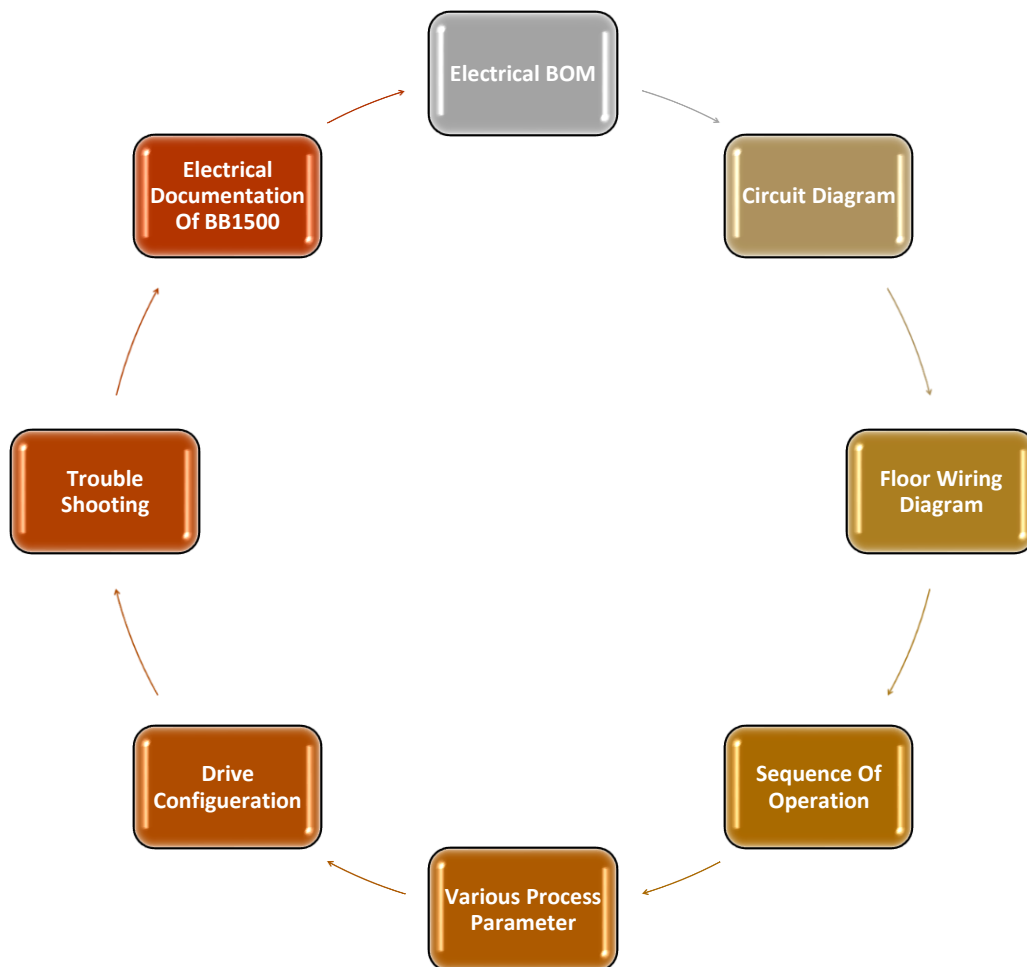
Duration & Place of the Training -

I undertook my internship training at **LohiaCorp**, a well-established industrial and manufacturing company known for its expertise in advanced machinery and automation systems. The company has its manufacturing unit located in **[Peenya Industrial State, Bengaluru]**, and is actively engaged in the production of high-performance machines used in textile and industrial sectors.

My internship began on **14th May 2025** and continued for a period of **5 weeks**. During this time, I was assigned to the **Electrical Department**, where I had the opportunity to work closely with experienced engineers and technical staff.

Purpose Of the Intership –

Throughout the internship period, I was exposed to real-time industrial practices involving **machine circuit analysis, component testing, electrical troubleshooting, and the use of instruments such as digital multimeters and tachometers.**



The environment at LohiaCorp Ltd provided me with a valuable hands-on learning experience and helped me bridge the gap between academic knowledge and practical application. This duration proved to be highly beneficial for enhancing my technical understanding and has contributed significantly to my personal and professional growth.

Organization Profile

Brief About the Company -

LohiaCorp Ltd. is a leading global manufacturer of machinery and equipment for the woven plastics industry. With decades of experience and innovation, the company has established a strong presence in both domestic and international markets. Lohia specializes in developing and delivering high-quality machinery used in the production of woven sacks, fabrics, and technical textiles, including tape extrusion lines, circular looms, and winders.

Founded with a commitment to engineering excellence and customer satisfaction, LohiaCorp Ltd has grown into a highly respected industrial brand with operations in over 85 countries. Its headquarters located in Kanpur, UttarPradesh and manufacturing units are located in Bengaluru as well, where a state-of-the-art facility supports R&D production, quality control, and testing.

What Company do-

LohiaCorp Ltd. is a leading industrial manufacturing company specializing in machinery and complete solutions for producing woven polypropylene (PP) and high-density polyethylene (HDPE) fabric products. These fabrics are essential across diverse sectors like packaging, agriculture, construction, and infrastructure.

The company designs and manufactures high-performance machines, including

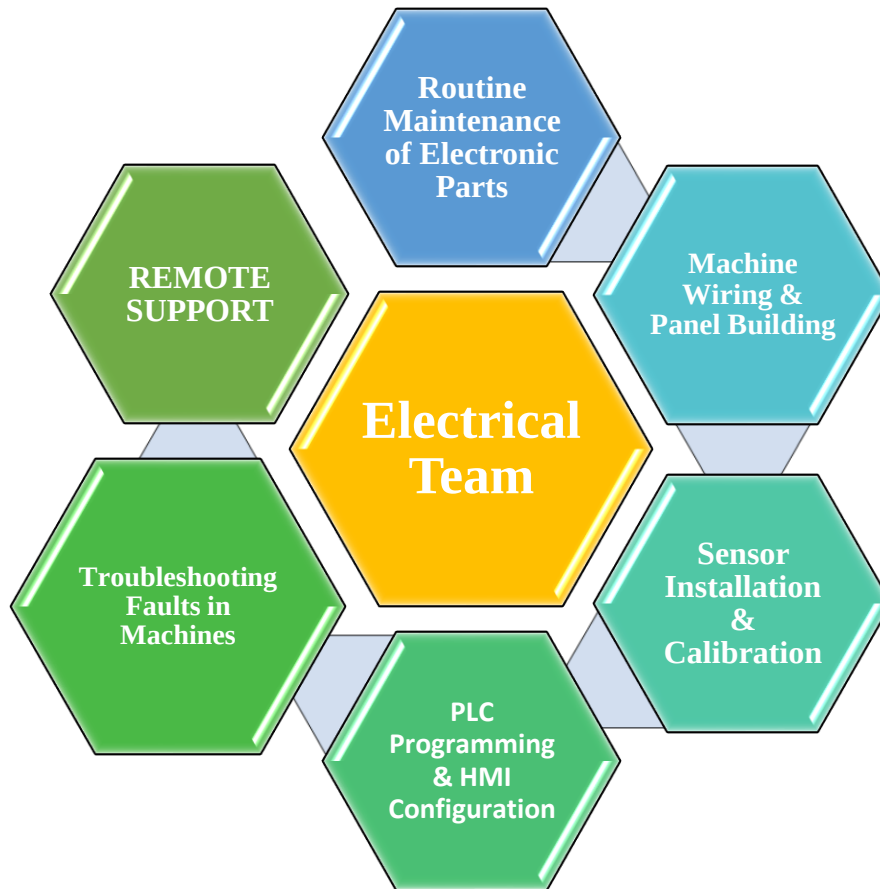
- **Tape extrusion lines for producing flat PP/HDPE tapes.**
- **Circular weaving machines (looms) for weaving tapes into tubular fabric.**
- **Winders and fabric processing equipment for handling, cutting, and finishing.**
- **Integrated automation and electronic control systems for precise operation.**

Beyond machinery, Lohia is committed to R&D, technical service, and customer training, offering comprehensive support and long-term value. Driven by innovation and quality, the



company continuously enhances its products to meet modern demands like energy efficiency and digital control, making Lohia a trusted global brand in woven plastics machinery.

Services Provided by Electrical Team at Lohia-



The Electrical Team at LohiaCorp Ltd plays a critical role in the entire lifecycle of industrial machinery, from initial setup to ongoing maintenance. Their expertise ensures machines operate efficiently, precisely, and reliably.

1. CONTROL PANEL FABRICATION & WIRING

The team designs, builds, and maintains control panels for a diverse range of machines, including Tape Lines, Circular Looms, and Winders. This involves intricate wiring and assembly to ensure all electrical components are properly integrated and protected.



2. PLC PROGRAMMING & AUTOMATION

Many of Lohia's machines rely on Programmable Logic Controller (PLC) systems for automated operations. The Electrical Team is responsible for installing and meticulously programming these PLCs, ensuring precise control over critical machine functions like cutting, winding, and tension management.

3. HMI (HUMAN MACHINE INTERFACE)

To facilitate intuitive machine operation, the department configures HMI screens. These interfaces allow operators to effortlessly monitor and control vital machine parameters in real-time, such as motor speed, temperature, product length, and fault status.

4. SERVO MOTOR AND DRIVE SYSTEM

For high-speed and high-precision applications, like Automatic Winders, the team handles the complete integration of Servo Motor and Drive Systems. This includes their installation, meticulous wiring, and precise feedback calibration using advanced sensors and encoders.

5. SENSORS AND FEEDBACK INTEGRATION

The Electrical Department is responsible for fitting and calibrating various sensors—including proximity, temperature, pressure, and load cells—onto machines. These sensors provide crucial feedback, enabling machines to react to environmental changes and maintain optimal performance.

6. TROUBLESHOOTING & MAINTAINANCE

When electrical issues arise, the Electrical Team acts swiftly. They diagnose faults using specialized tools like multimeters and testers, effectively troubleshooting and resolving issues to minimize machine downtime and ensure continuous operation.



7. REMOTE SUPPORT

Lohia Corp has world class service support to their end users. Their remote maintenance modules allow their customer technical support team to erect and troubleshoot entire machine online.

This department is a **backbone for machine performance** at Lohia. As an intern, I had the chance to observe how electronic systems work together with mechanical parts, giving me real-time experience in the field of industrial Electrical.

Weekly Work Summary

Components Handled -

My internship at **LohiaCorp Ltd** provided an invaluable opportunity to immerse myself in the world of industrial machinery, specifically focusing on its intricate **electrical and electronic components**. I actively **worked with and observed** a comprehensive range of these critical elements, understanding their vital roles in **machine control, automation, protection, and signal processing**. Here's a breakdown of the major components I handled or analyzed, detailing their core functions and purposes

Components	Uses
Connected Wires	Connected wires are two or more wires joined to allow current to flow between them.
Non Connected Wires	Non-connected wires are wires that cross each other without any electrical connection.
SPST Toggle Switch	SPST (Single Pole Single Throw) Toggle Switch is a manual switch that opens or closes a single circuit using a toggle lever.
SPDT Toggle Switch	SPDT (Single Pole Double Throw) Toggle Switch is a switch with one input and two output terminals, operated by a toggle lever.
Push Button Switch (N.O)	A Push Button Switch (N.O) is a normally open switch that closes the circuit only when pressed.
Push Button Switch (NC)	A Push Button Switch (NC) is a normally closed switch that opens the circuit when pressed.
DIP Switch	A DIP (Dual In-line Package)Switch is a set of small manual switches packaged in a dual in-line form.
SPST Relay	An SPST relay is an electrically operated switch with a single pole and single throw contact.
Variable Resistor(Rheostat)	A variable resistor (rheostat) is a resistor whose resistance can be manually adjusted.
Iron Core Inductor	An iron core inductor is a coil wound around an iron core to increase its inductance.

Voltage Source	A voltage source is a device or component that provides a constant voltage.
Current Source	A current source is a device that delivers a constant current regardless of voltage changes.
AC Voltage Source	An AC voltage source provides an alternating voltage that changes polarity and magnitude over time.
Controlled Voltage Source	A controlled voltage source is a voltage source whose output depends on another voltage or current in the circuit.
Controlled Current Source	A controlled current source is a current source whose output depends on another voltage or current in the circuit.
Low Frequency AC Generator	Generator that produces low-frequency alternating current (usually below 50 Hz).
Medium Frequency AC Generator	Generator that produces AC in the medium frequency range (typically 400–10,000 Hz).
High Frequency AC Generator	Produces high-frequency alternating current (above 10 kHz).
Ideal Current Generator	A theoretical source that supplies constant current regardless of load.
Ideal Voltage Generator	A theoretical voltage source that maintains constant voltage regardless of load.
Dynamo With Built in Breaks	A mechanical-to-electrical generator with integrated braking system.
Homopolar DC Generator	A generator that produces direct current using a rotating disc and magnetic field.
Pulse Generator	A circuit that generates electrical pulses.
Tachometer Generator	A generator that produces voltage proportional to rotational speed.
3 Phase Synchronous generator with WYE Star Connection	A 3-phase generator with star-shaped (WYE) winding configuration.
Fuse Switch	A switch combined with a fuse for circuit protection.

Isolator Switch Disconnecter	A switch that disconnects a part of a circuit for safe maintenance.
Fuse Switch Disconnecter	A combined fuse and isolator switch unit.
Protection Resistor	A resistor used to limit current or voltage surges.
Fast Blow Fuse	A fuse that blows quickly on overcurrent.
Slow Blow Fuse	A fuse designed to tolerate brief surges before blowing.
Fuse With Striker	A fuse equipped with a striker mechanism that triggers on blowing.
Fuse With Alarm Contact	A fuse that can send an alert signal when blown.
Fuse with Separate Alarm Contact	A fuse with an independent contact for alarm signaling.
3 linked Fuse With Release of Any Striker	Three fuses linked together to trip if any one blows.
Non Draw Out Circuit Breaker	A fixed circuit breaker that cannot be physically removed from its enclosure without disconnecting wiring.
Draw Out Circuit Breaker	A circuit breaker that can be physically withdrawn for maintenance or replacement.
Network Protector	A device that protects electrical networks from faults and backfeed.
Threaded Stud Type Circuit Breaker	Circuit breaker with threaded stud terminals for secure cable connections.
Double Pole Circuit Breaker	A breaker that interrupts two conductors simultaneously.
Three Pole Circuit Breaker	A breaker that interrupts all three phases in a three-phase system.
Isolator Circuit Breaker	A breaker that acts as both an isolator and a protective device.

Lightening Arrester	A device that protects electrical equipment by diverting lightning surges to ground.
Spark Gap	A gap between two electrodes that allows spark discharge when voltage exceeds a threshold.
Double Spark Gsp	Two spark gaps in series or parallel for enhanced voltage protection.
Electrical Motor Winding Coil	Wire coils in a motor where current creates magnetic fields.
Carbon Brush	A conductive block that transfers current between stationary and rotating parts of a motor.
Generic Motor	A general term for any electrical motor converting electrical energy to mechanical energy.
Dual Speed Motor	A motor designed to operate at two different speeds.
AC Motor	Motor powered by alternating current.
DC Motor	Motor powered by direct current.
Linear Motor	A motor that produces linear motion instead of rotational.
Stepper Motor	A motor that moves in discrete steps or increments.
Permanent Magnet DC Motor	A DC motor with permanent magnets instead of field windings.
AC Single Phase Series Motor	A single-phase AC motor with series winding for high starting torque.
DC Series Motor	A DC motor with field winding in series with the armature.
Single Phase Repulsion Motor	A single-phase motor with repulsion start mechanism.
DC Shunt Motor	DC motor with field winding connected in parallel with the armature.

Single Phase Synchronous Motor	A single-phase motor that runs at synchronous speed.
DC Compound Excitation Motor	A DC motor with both series and shunt field windings.
3 Phase AC Motor	A motor powered by three-phase AC supply.
3 Phase star Shaped Motor	A 3-phase motor with star (WYE) connected windings.
3 Phase Wound Rotor Motor	A motor with rotor windings connected through slip rings.
3 Phase Linear Motor	A 3-phase motor producing linear motion.
General Switch Symbol	A schematic symbol representing a generic switch.
Open Switch	A switch in the OFF or disconnected position.
Closed Switch	A switch in the ON or connected position.
Thermal Magnetic switch	A switch that trips on thermal overload or magnetic short circuit.
Limit Switch	A switch activated by physical movement or position of an object.
Differential Switch	A switch that operates based on differences in electrical parameters like current.
Pulse Counter Switch	A switch that counts electrical pulses each time it is activated.
Switch SPDT	A switch that connects one input to one of two outputs.
Slide Switch SPDT	A sliding version of SPDT switch.
Switch DPST	A switch that controls two circuits simultaneously with ON/OFF function.

Switch DPST One close Before Other	A DPST switch where one pole closes before the other.
Switch DPDT	A switch controlling two circuits with two possible output paths each.
Switch SP4T	A switch connecting one input to one of four outputs.
Rotary Switch	A switch operated by rotating a knob to select circuits.
On Delay Switch	A switch that delays turning ON after being activated.
On Off Delay Switch	A switch that delays both ON and OFF actions.
NO Push Button	A push button that is open until pressed.
NC Push Button	A push button that is closed until pressed.
Push Button Lockable	A push button that can be locked in the ON position.
Positive Opening Push Button Mushroom Head	A large mushroom-shaped emergency stop push button that positively opens the circuit.
Positive Make Contact Push Button	A push button that positively ensures contact closure when pressed.
Timer Switch	A switch that operates after a preset time delay.
Delayed Closing Switch	A switch that closes after a delay when activated.
Fast Closing Switch	A switch designed to close very quickly.
Delayed Opening Switch	A switch that opens after a preset delay once activated.
Fast Opening Switch	A switch designed to open quickly.

Push To Make Switch	A normally open push button that closes the circuit only when pressed.
Push To Break Switch	A normally closed push button that opens the circuit when pressed.
Push Button Limit Switch	A push button activated by mechanical movement to limit motion.
Push Button Double Pole Limit Switch	A limit switch controlling two circuits simultaneously.
Push Button SPDT	A push button switch with Single Pole Double Throw contacts.
Push Button DPDT	A push button switch with Double Pole Double Throw contacts.
Float Switch	A switch activated by the level of liquid in a tank.
Thermal Switch	A temperature-activated switch.
NC Thermal Switch (Normally Closed)	A thermal switch that stays closed until a certain temperature is reached.
NO Thermal Switch (Normally Open)	A thermal switch that stays open until a certain temperature is reached.
Selector Switch	A manual switch to select between multiple circuits.
Electronic Fluorescent Starter	An electronic device to start fluorescent lamps.
Switch DPST BBM	Double Pole Single Throw switch with “Break Before Make” action.
Proximity Switch	A switch activated by the presence of a nearby object without contact.
Red Cable	To provide 220v to 24v AC Current in the Machine
Blue Cable	To provide 24v DC Current in the Machine



Light Blue Cable	It is Neutral In the Circuit
Black Cable	To show the AC/DC Power In the Circuit
Yellow/Green Cable	PE(Power/Protective Earth)
Green Cable	IE (Instrumentation Earth)

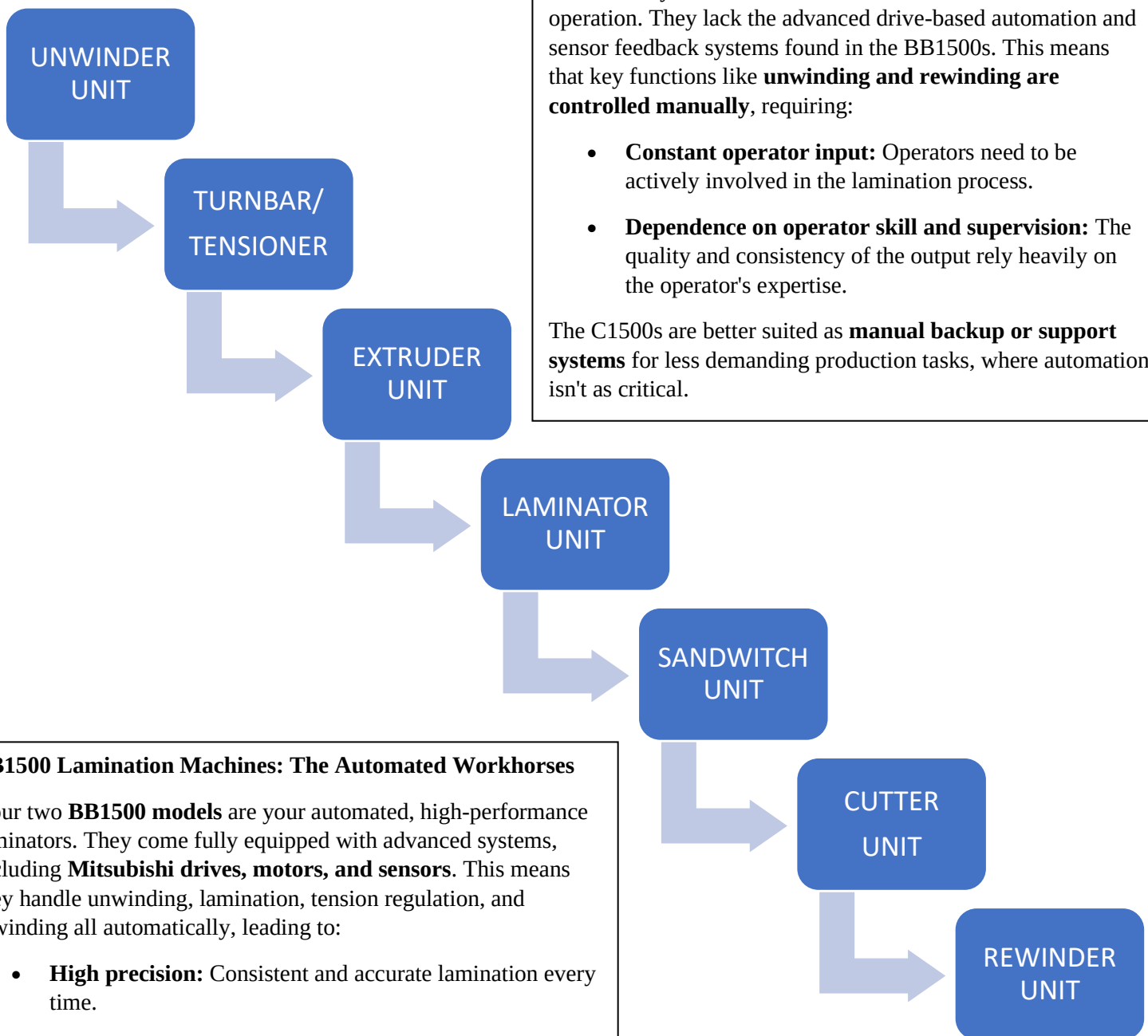
Machine Observed (LAMINATION)

C1500 Lamination Machines: The Manual Backup

In contrast, your two **C1500 machines** are more manual in their operation. They lack the advanced drive-based automation and sensor feedback systems found in the BB1500s. This means that key functions like **unwinding and rewinding are controlled manually**, requiring:

- **Constant operator input:** Operators need to be actively involved in the lamination process.
- **Dependence on operator skill and supervision:** The quality and consistency of the output rely heavily on the operator's expertise.

The C1500s are better suited as **manual backup or support systems** for less demanding production tasks, where automation isn't as critical.



BB1500 Lamination Machines: The Automated Workhorses

Your two **BB1500 models** are your automated, high-performance laminators. They come fully equipped with advanced systems, including **Mitsubishi drives, motors, and sensors**. This means they handle unwinding, lamination, tension regulation, and rewinding all automatically, leading to:

- **High precision:** Consistent and accurate lamination every time.
- **Consistent output:** Reliable quality across large production runs.
- **Reduced manual intervention:** Less need for operators to constantly monitor and adjust the machine.

These machines are ideal for **high-speed, high-volume production** where efficiency and consistency are key.

Observation Over BB1500 Lamination Machine

Introduction to the machine process –

The BB1500 Lamination Machine is a high-performance industrial system that precisely laminates multiple film or sheet layers. It's often used for flexible packaging, industrial tapes, and coated substrates. Built with **Mitsubishi drives, AC motors, and industrial-grade sensors**, it offers both speed and accuracy for continuous production.

How it Works –

The BB1500 bonds materials using **heat and pressure** through a synchronized series of motor-driven rollers, heaters, and tension-control systems. This process ensures optimal adhesion and smooth surfaces while preserving the laminated product's mechanical and aesthetic qualities.

Automation and Control –

Mitsubishi Variable Frequency Drives (VFDs) manage the machine's automation and control by regulating motor speeds, torque, and direction. **Integrated sensors** provide real-time feedback on web position, temperature, pressure, and tension, allowing for precise adjustments and consistent product quality.

Production Stages and Safety –

From unwinding raw material rolls to laminating through heated nip rollers and finally rewinding the laminated sheet, each stage of the BB1500 is carefully synchronized. The system also includes **safety interlocks, emergency stops, and diagnostic feedback** for reliable and safe operation.

Advanced Capabilities –

The BB1500 lamination machine is an advanced platform that supports **high-throughput manufacturing**. It also offers flexibility in adjusting critical parameters like speed, pressure, and temperature to meet the specific requirements of various lamination jobs.

- **Summary Of the Working of Machine BB1500-**

Unwinding and Tensioning (The Initial Setup)-

The journey begins with the unwinder, a crucial component that carefully unrolls the raw material from its large supply roll. This initial unwinding sets the stage for the entire process. Immediately following, a hydraulic press unit takes over, applying precise tension to the material. This ensures the material remains taut and stable, preventing wrinkles or misalignments that could compromise the final product. A strategically placed sensor then guides the roller, ensuring the raw material is consistently positioned at the optimal location for subsequent stages.

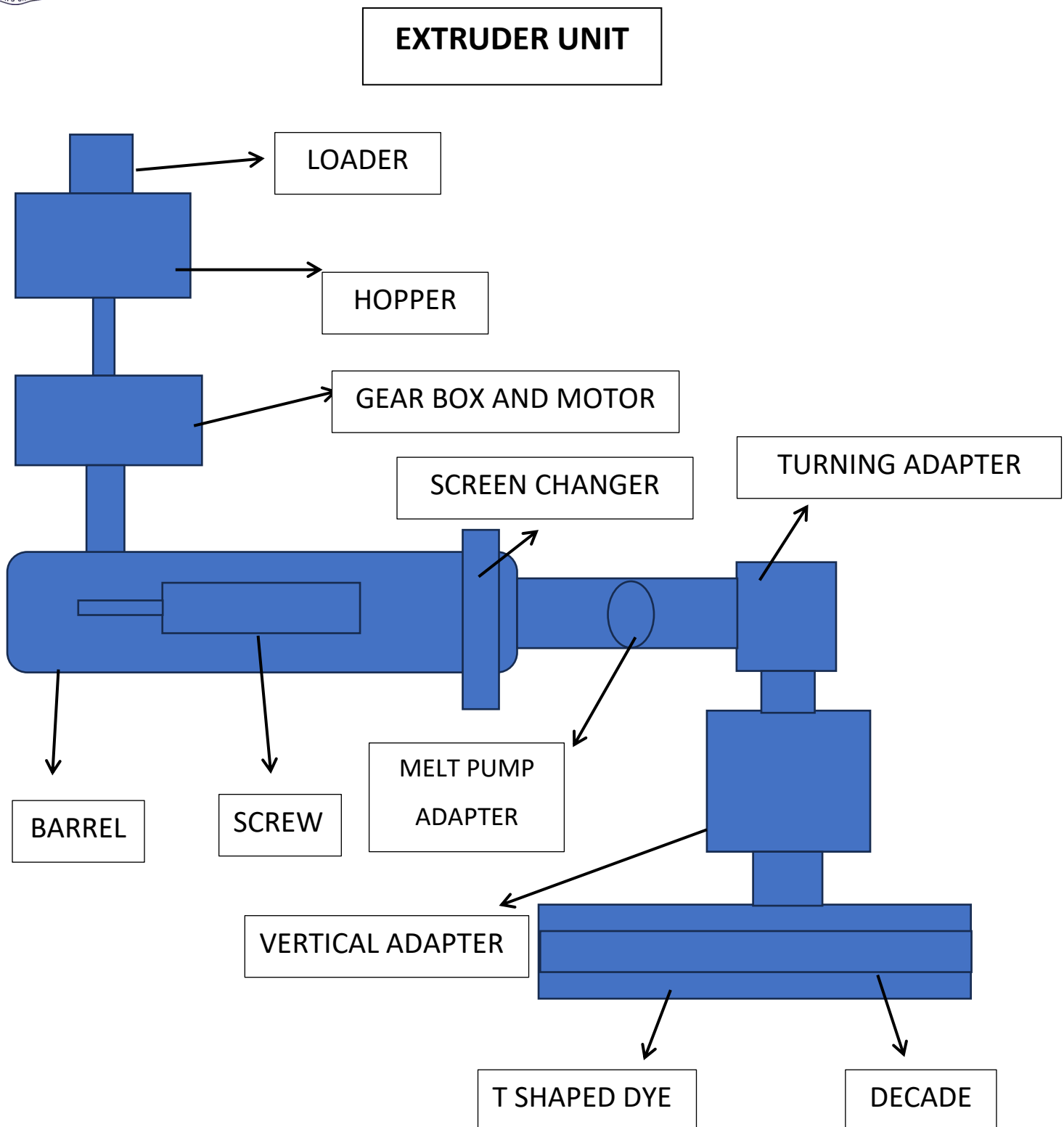


In the Unwinder Unit there are Breakers are present which helps to provide the required tension to the material Initially it don't count the tension but after few seconds when the sensor detect that the Material is loosen then it got tensionised by the range of +2 or -2 difference. For eg if we give 200 it will vary from 198 newton to 202 Newton.

To prepare the material for superior adhesion, an oil barrel preheats it. This controlled heating process enhances the material's receptiveness to coatings, ensuring a strong and lasting bond.

LOAD CELL- It is a device which measures the tension of the fabric if the Value of the tension of the fabric is not to much then it provides the feedback to the controller to the tensioner which then increases the tension as per the Requirement and as per given by the load cell. It works on the Principle of **Piezo Electric Effect.**





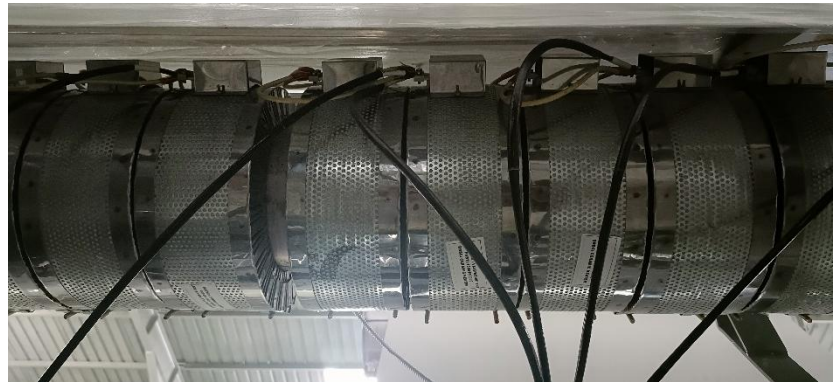
The prepped raw material then moves towards the heart of the lamination process. This involves a series of interconnected units:

Loader and Hopper: These components facilitate the feeding of the raw material into the system.

Gear Box and Motor: This provides the necessary power and control for the extrusion process.

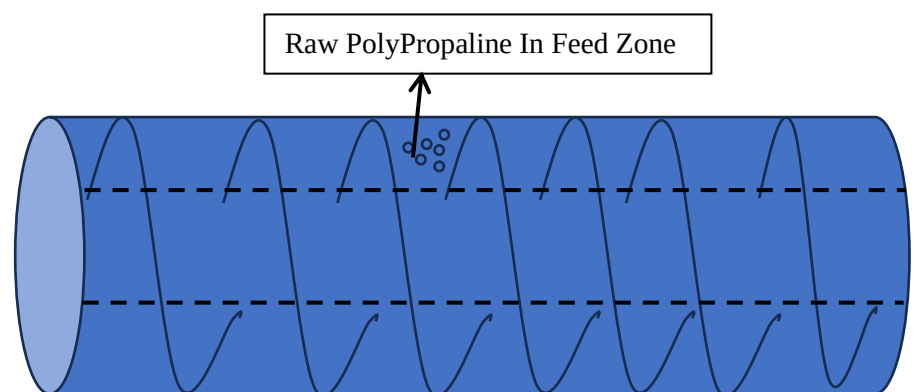
Screen Changer and Turning Adapter: These units likely manage the filtration of the molten material and direct its flow.

Barrel and Screw: Here, the material is subjected to heat and pressure, transforming it into a molten state. The screw, a rotating helix, efficiently conveys and melts the material. Screw consist of three zones which helps to provide the melt of polypropaline which are-



1. Feed Zone
2. Compressor Zone
3. Melting Zone

Each zone has it's own unique Properties Feed Zone Provides space where polypropaline is stored in the screw. Compressor zone Provides compression which allows to compress the molten polypropeline with heat and pressure, Melting Zone allows to make it completely in molted liquid Form.



SCREW REPRESENTATION IN BARREL

Melt Pump Adapter and Vertical Adapter: These units guide the molten material towards the final coating stage.

Enhancing Adhesion and Applying the Coating is A critical step in achieving strong lamination is the Corona Treater. This sophisticated device enhances both the mechanical and chemical adhesion properties of the raw material. It works by emitting electrons onto the

material's surface, creating microscopic pores and increasing its surface energy. This allows the subsequent coating material to bond more effectively.

Corona Treator is present in the machine it is basically a device which is use to enhance the adhesion of the material (Mechanically and Chemically). It use to emit electrons to the surface of raw material which allows to provide the pores and adhesion to the material then the dye releases the coating material which directly falls on the fabric which then go through the chill roller to make it chill and stable. The extruder unit is a moving unit which get moves and comes on top of the NIPROLL Which provides the Lamination adhesion.



Laminator Unit-

This unit helps Comprises of one rubber and one chill roller when the extruder unit comes in top of it and the molten polypropaline falls on the fabric we requires a rubber and a chill roller.

Rubber Roller- It helps to give a impressional support to the fabric by the help of which fabric gets laminated nicely and without the wrinkles on it.

Chill Roller- It helps to make the laminated fabric instant cool by this process the lamination will be more adhesive and there will be no chance of the fabric to loose the laminated film of polypropaline from it.



A T-shaped dye precisely releases the molten coating material, which then flows directly onto the prepared fabric. To solidify and stabilize this fresh coating, the fabric immediately passes through a chill roller, rapidly cooling the material and ensuring a smooth, uniform finish. In BOPP there is one unwinder is present which unroll the roll of BOPP.

Precision Cutting and Multi-Layer Lamination-

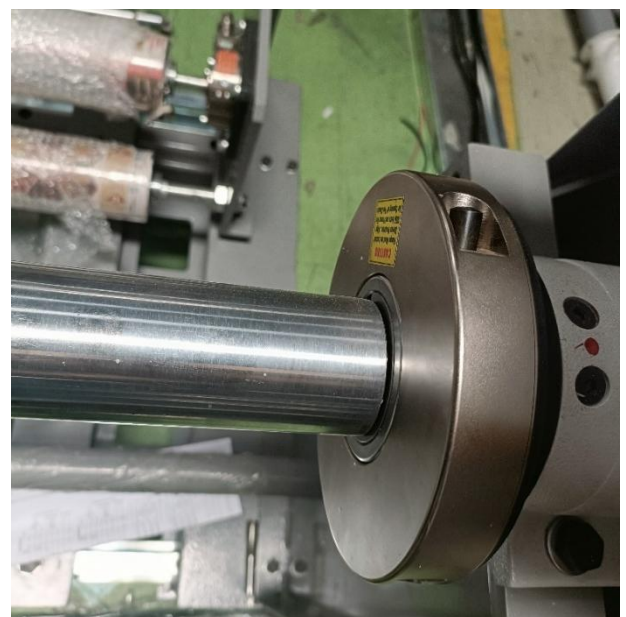
After the lamination, the material moves to an actuator, equipped with a light sensor. This sensor accurately detects the material's position, allowing the actuator to precisely cut the fabric to the desired length. The actuator helps in the cutting process we just need to set it once and all other things will be done automatically.

A unique aspect of this process is the use of double-layered material. To ensure complete coverage and integrity, an additional three inches of polypropylene are added. This polypropylene is then precisely cut by the actuator and subsequently moves to the BOPP layer.



BOPP Lamination and Branding-The BOPP (Biaxially Oriented Polypropylene) layer is where branding and advertising come to life. This section is responsible for printing a thin, often decorative, layer onto the fabric, crucial for product identification and marketing in the laminator.

Within the BOPP unit, an unwinder is present to meticulously unroll the BOPP film. This unwinder unit incorporates brakes that provide the exact tension needed for the BOPP material. Initially, the system might not actively measure tension, but after a few seconds, once a sensor detects any looseness in the material, it adjusts the tension within a precise range of ± 2 Newtons. For example, if a target tension of 200 Newtons is set, the system will maintain it between 198 and 202 Newtons, ensuring consistent application.



The Fabric compensator plays A crucial role of arranging the BOPP Filmed material across the operation of machine.

The BOPP layer is only laminated when the polypropylene is completely melted. If not, the lamination will be imperfect. The BOPP unit is equipped with two unrollers, enabling printing on both sides of the raw material. The process involves the fabric passing through, one printed side returning through rotations, and then the unprinted side being presented for printing on the other side.

Alignment and Finishing Touches-

Precise alignment is critical throughout the process. Two sensors play a vital role here:-

Floating Edging Surface: This sensor, part of the cutting unit, refines the edges of the material and ensures the sheet remains perfectly centered on the roll as it moves. It's designed to move the entire body of the cutting unit for comprehensive alignment.

Web Aligner: This sensor also contributes to material alignment but operates by moving only from the top, fine-tuning the material's position. The main difference between the web aligner and floating edging surface is the movement floating edge will move the whole body for the fabric but the web aligner will only move it's top by the moving of top the fabric gets aligned.

Perforation Unit-

After the complete lamination process, perforation occurs. This involves adding micro-holes to the material, facilitating air ventilation and enhancing the product's breathability or other functional properties.



The perforation is very Important for the lamination machine if we don't do the perforation the bags material will be wasted if any kind of air is trapped inside of it. By creating micro holes it will help in the refining of the material and reduces the chances of creating a damaged product.

REWINDER UNIT-

This is the last step of the machine which consists of many proximity sensors and Pneumatic. Some Major Parts of the rewinder are as follows-

1. SWIVEL ARM
2. HYDRAULIC ARM
3. CUTTER



Swivel Arm- This helps to perform the operation of taking the rolled fabric to the handover position and it can be performed as UP and DOWN which helps to maintain the stability in the machine.

Hydraulic Arm- After the swivel arm operation it get handover to the hydraulic arm which disengage through the clamp operation and the roll get extracted from the machine.

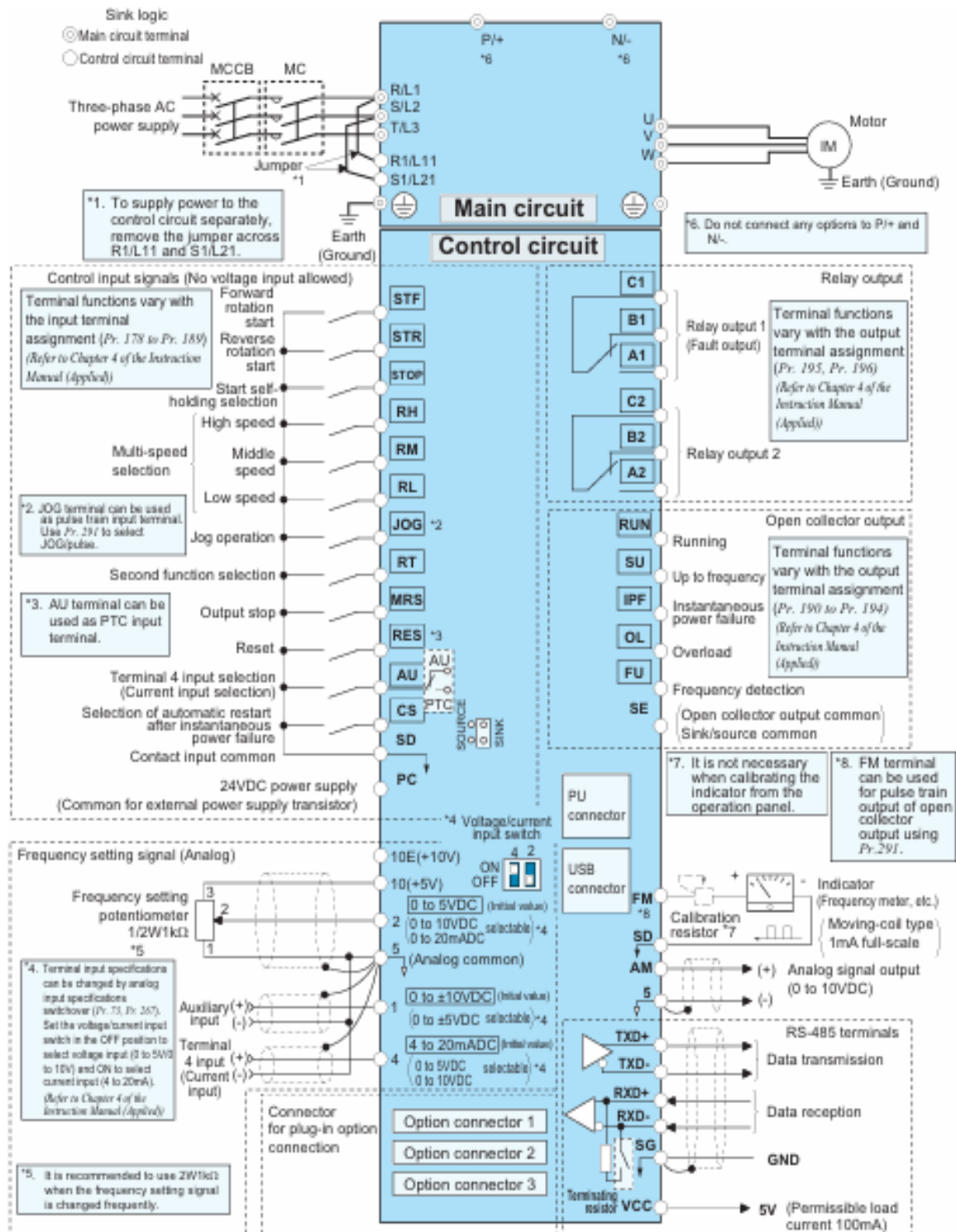


Cutter- In the Rewinder One pneumatic cutter is present which only helps to cut the fabric at the end of machine. It works on the Pneumatic air Operation and cuts the fabric after the cutting it gets to the handover Position.

CIRCUITS TESTED OF BB1500 –

MITSUBISHI DRIVE-

In our machine we used the mitsubishi drive to controls all the motors one by one each motors consist of the same drive In our cases there are 7 CORE MOTORS are present so we are using total 7 Number of drives.



IMPORTANT PINS OF MITSUBISHI DRIVE FOR BB1500-

- PIN NUMBER 2 – **Frequency setting (voltage)** Inputting 0 to 5VDC (or 0 to 10V, 0 to 20mA) provides the maximum output frequency at 5V (10V, 20mA) and makes input and output proportional. Use Pr. 73 to switch from among input 0 to 5VDC (initial setting), 0 to 10VDC, and 0 to 20mA. Set the voltage/current input switch in the ON position to select current input (0 to 20mA).
- PIN NUMBER 5 – **Frequency setting common** Common terminal for frequency setting signal (terminal 2, 1 or 4) and analog output terminal AM. Do not earth (ground).
- PIN NUMBER S1 – **Digital Input Signal 1** Often used to receive a control signal, such as **Start, Forward, or ON** command from a switch, PLC, or relay. In our case it for emergency s1 s2 and pc will be looped together and in emergency condition they will stop the machine.
- PIN NUMBER S2 – **Digital Input Signal 2** Used for a second control signal, commonly for **Stop, Reverse, or OFF** function, depending on configuration. In our case it for emergency s1 s2 and pc will be looped together and in emergency condition they will stop the machine.
- PIN NUMBER PC – **Positive Common (for digital inputs)** Acts as a **reference voltage** (often +24V DC) for digital inputs like S1, S2, etc. Inputs like S1 and S2 are active when connected to **PC** through external switches or logic signals. In our case it for emergency s1 s2 and pc will be looped together and in emergency condition they will stop the machine.
- PIN NUMBER DC+ It provides the **positive terminal** of the required **DC voltage** to the motor.
- PIN NUMBER DC- It provides the **negative terminal** (ground/return path) of the **DC voltage** to the motor.
- PIN NUMBER RES – It is very Important because it controls the alarm

- PIN NUMBER STF – **Forward Rotation Start** Turn ON the STF signal to start forward rotation and turn it OFF to stop.
- PIN NUMBER STR – **Reverse rotation start** Turn ON the STR signal to start reverse rotation and turn it OFF to stop.
- PIN NUMBER SD – **Contact input common (sink) (initial setting)** Common terminal for contact input terminal (sink logic) and terminal FM.

External transistor common (source) When connecting the transistor output (open collector output), such as a programmable controller, when source logic is selected, connect the external power supply common for transistor output to this terminal to prevent a malfunction caused by undesirable currents.

BASIC CONNECTIONS MADE THROUGHOUT THE INTERNSHIP PERIOD

• RELAY MODULE OF UNWINDER –

A relay module integrates one or more electromechanical relays with supporting circuitry, enabling it to function as an electronically controlled switch.

WORKING –

Relays operate on the principle of electromagnetism. When an electric current energizes the relay's coil, it generates a magnetic field. This magnetic field then actuates a movable switch, known as an armature, which either

opens or closes an electrical circuit.

Types of Relay Modules:

Followings are the types of basic relay which we use in our machines.

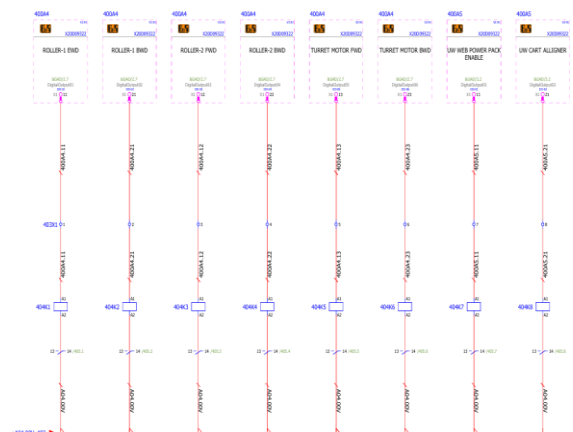
Single-Channel Relay Modules – For One device.

Multi-Channel Modules: Available in configurations like 2-channel, 4-channel, and 8-channel.

Solid-State Relay (SSR) Modules: Feature no moving parts, offering faster switching and quieter operation.

• UNWINDER CART POWER PACK –

The **Unwinder CART Power Pack** in our machine is essential for precisely feeding material—like paper, foil, or fabric—from a large roll into the subsequent processing sections. This integrated system ensures smooth, controlled unwinding.



It's a robust combination of a **motor, gearbox, brake, and a control panel**, all working in harmony to provide precise rotation, meticulous tension control, and reliable braking.

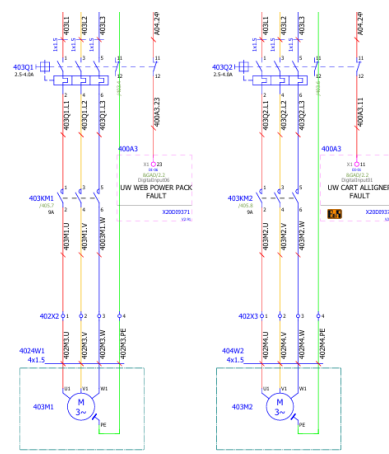
FEATURES –

Gentle Start: The **VFD** initiates the motor with a smooth, jolt-free power delivery as the machine starts.

Continuous Feed: The roll begins its steady rotation, releasing material into the process.

Dynamic Tension Adjustment: Should material tension fluctuate, the **tension sensor** instantaneously triggers adjustments to the motor speed or brake engagement, keeping everything perfectly arranged.

Controlled Shutdown: Operators can easily stop the motor or engage the brake—either manually or through automated systems—as the roll approaches its end.



• WEB ALIGNER POWER PACK –

The **Web Aligner Power Pack** is a critical component in machines that handle continuous roll materials like paper, film, foil, or fabric. Its purpose is to maintain the material's precise lateral position as it moves.

As part of a **Web Guiding System**, its primary function is to automatically correct any side-to-side shifting (left-right movement) of the material roll, thereby preventing misalignment.



• CONTACTORS –

A **Contactor** is basically a **heavy-duty switch** that is used to **control the ON/OFF of motors, heaters, pumps, lights**, or any high-power load **automatically or remotely** using a small control signal.

HOW TO USE?

- ☐ We press **Start push button**.

- ☐ **Control circuit energizes the coil** inside the contactor.
- ☐ Coil pulls the **main contacts together** → Current flows to motor → Motor starts.
- ☐ Press **Stop button**, coil de-energizes, contacts open → Motor stops.

FEATURES OF CONTACTORS –

- Can control **high current with a small control signal**
- **Remote control** of machines
- **Long life** if properly used
- Can be combined with **timers, relays, sensors** for automation



SMPS FOR MODULE –

A **Switched Mode Power Supply (SMPS)** is a highly efficient, compact power supply that converts standard 230V AC mains electricity into stable, low-voltage DC outputs such as 5V, 12V, or 24V. We utilize these DC outputs to reliably power control modules, relay modules, PLCs, sensors, Arduino boards, and various other low-voltage electronic devices.

WORKING –

- ☐ **AC Input:** It takes the 230V AC input from the mains.
- ☐ **High-Frequency Switching:** This AC is then converted to DC through a high-frequency switching process.
- ☐ **Stable DC Output:** The result is a regulated and stable DC output (e.g., 24V, 12V, or 5V) available at its output terminals.



SMPS FOR UNWINDER CONTROLLER –

In our unwinder system, the **SMPS (Switched Mode Power Supply)** is critical for delivering stable DC power to the **Unwinder Controller**. This controller typically requires 24V DC to function correctly, and that's precisely what the SMPS provides.

The unwinder controller is the brain of the system, managing crucial functions such as:

- **Tension control**
- **Brake logic**
- **Roll speed control**
- **Feedback from sensors** (like load cells, encoders, etc.)

All these operations demand a clean and stable 24V DC power supply. Since we can't directly power them from AC mains, we install an SMPS between the AC supply and the controller.



SMPS FOR MATCHING BOPP –

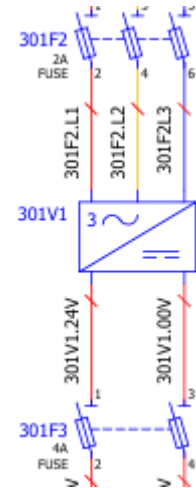
In **BOPP film machines**, particularly within the **matching or edge alignment unit**, the **SMPS (Switched Mode Power Supply)** plays a vital role. It delivers a stable 24V DC or 12V DC power supply to critical components such as:

- **Web Guide Controllers**
- **Sensors** (Proximity / IR)
- **Linear Actuators / Motors**
- **Relay Modules**
- **Signal Converters**



The primary objective of web alignment (or matching) systems in BOPP film production is to maintain the precise lateral positioning of the film. The highly sensitive sensors and controllers employed for this task demand a clean, stable, low-voltage DC supply.

Since direct 230V AC supply is unsuitable for these electronic components, the SMPS efficiently converts the 230V AC mains power into the required 24V DC (or 12V DC).



FABRIC COMPENSATOR –

A **Fabric Compensator**, also known as a **dancer unit** or **web compensator**, is a crucial mechanical system designed to maintain optimal tension in fabric or web materials during processes like lamination. It effectively absorbs slack, compensates for speed variations, and ensures smooth, wrinkle-free material handling.

Within the **matching system** of the BOPP line, this unit is commonly referred to as the **Web Guide Controller** or **Edge Guiding Controller**. The system allows fabric to move in forward or reverse direction so that print on BOPP can be matched.



The controller's primary role is to ensure the film travels perfectly straight by:

- **Interpreting signals** from the edge sensor.
- **Sending output commands** to the actuator or guide frame.
- **Maintaining accurate edge positioning**, preventing side shifting or misalignment.

SMPS FOR CONTROLLER –

In our BOPP and other web alignment machines, utilizing **controllers** necessitates a clean and stable DC power supply. For this crucial requirement, we rely on a **Switched Mode Power Supply (SMPS)**.

Robust and precise E+L controllers, such as the **ELS 2030, DO 3010, and AGS series**, are engineered to operate on a consistent **24V DC** supply. They are the backbone of your alignment system, controlling:

- Accurate edge guiding
- Real-time sensor feedback interpretation
- Responsive actuator control



Attempting to power these advanced units directly with 230V AC is not only unsafe but also detrimental to their performance. The SMPS acts as the vital intermediary, delivering the clean, regulated 24V DC that ensures your E+L controller operates with maximum stability and precision.

- **JUNCTION BOX OF REWINDER (OPERATOR SIDE)–**

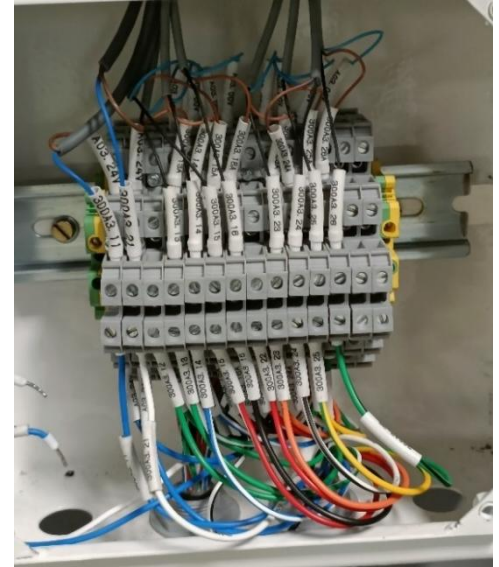
The **junction box** situated on the operator side of our rewinder unit serves as a compact electrical enclosure. It is the central consolidation point for all control and sensor wiring related to the rewinding process.

This vital hub provides a dedicated connection point for:

- **Limit switches**
- **Sensors & Indication Lamps**
- **Push buttons**
- **Brake or clutch feedback mechanisms**
- **Tension control signals**

PURPOSE OF JUNCTION BOX –

- ❑ **Enhanced Safety:** Provides a secure and enclosed environment for all operator-side device wiring.
- ❑ **Simplified Maintenance:** Facilitates local troubleshooting and repairs without needing to access the main control panel.
- ❑ **Expedited Troubleshooting:** Centralized connections make it easier to identify and resolve wiring issues.
- ❑ **Improved Organization:** Keeps complex field wiring neat, orderly, and protected.
- ❑ **Seamless Integration:** Acts as the critical interface between field devices and the main control panel.



• **HYDRAULIC MOTOR FOR REWINDER –**

In a rewinder system, the **hydraulic motor** is a critical component responsible for providing the precise and powerful rotational force needed to control the winding of materials onto a core. Unlike electric motors, hydraulic motors offer distinct advantages for demanding rewinding applications.

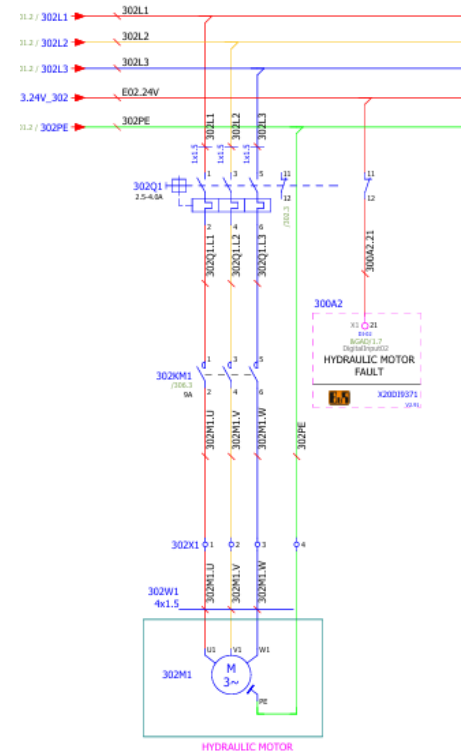
A hydraulic motor converts hydraulic fluid pressure into mechanical rotation. In a rewinder, this works as follows:

1. **Hydraulic Power Unit:** A separate hydraulic power unit (HPU) generates pressurized hydraulic fluid.
2. **Fluid Delivery:** This pressurized fluid is directed to the hydraulic motor via hoses and control valves.



- ## FUNCTIONS OF HYDRAULIC MOTOR –

- ❑ **Precise Torque Control:** Hydraulic motors excel at delivering consistent torque across a wide speed range, crucial for maintaining optimal tension during winding.
- ❑ **Variable Speed Operation:** They offer smooth and stepless speed adjustment, allowing for fine-tuning of the winding speed to match material requirements.
- ❑ **High Power Density:** Hydraulic motors can generate significant power from a relatively compact size, making them ideal for heavy-duty rewinding applications where space might be limited.
- ❑ **Smooth Acceleration/Deceleration:** The inherent properties of hydraulic systems allow for very smooth starts, stops, and changes in speed, preventing jerks or damage to delicate materials.
- ❑ **Holding Capability:** They can maintain a constant holding torque, essential for keeping the roll secure when the machine is stopped.



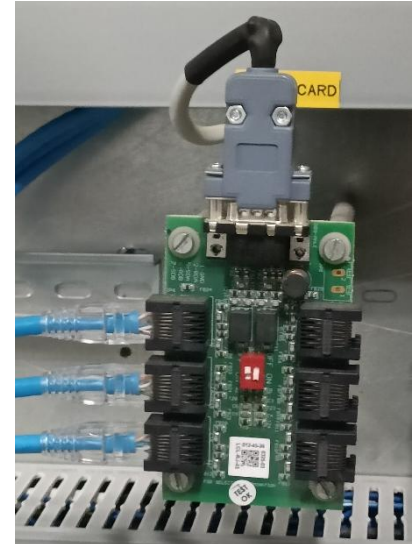
RJ 45 Card to RS485 – Here are a few refined options for your text about using RJ45 connectors for RS485 communication, with variations in tone and emphasis.

Streamlining RS485: Using RJ45 Connectors in BB1500 Machines –

In our BB1500 machines, we often leverage **RJ45 connectors** (commonly found on CAT5/CAT6 LAN cables) for **RS485 communication lines**. This innovative approach makes wiring controllers, VFDs, and sensors easier, enables longer-distance runs, and ensures a neater, more organized setup.

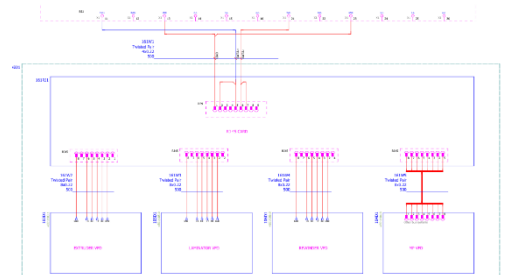
RS485: This is a robust standard for serial communication, widely used for data exchange between industrial devices like:

- **Tension controllers**
- **Load cell amplifiers**
- **Drives/VFDs (Variable Frequency Drives)**
- **HMIs (Human-Machine Interfaces)**



□ **RJ45:** This is a standard type of electrical connector, identical to those used for Ethernet (LAN) cables.

□ **RJ45-to-RS485:** This simply means we're using the physical RJ45 connector and a LAN cable to transmit RS485 signals between devices. We're utilizing the convenient cabling, not the Ethernet protocol itself.

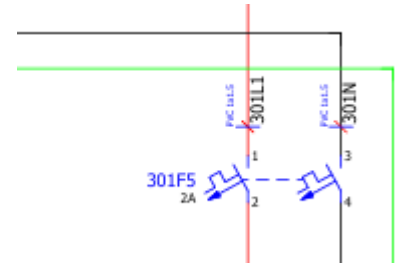


• **MCB POWER SUPPLY (MAIN POWER SUPPLY)** –

In our machine panels, the **MCB (Miniature Circuit Breaker)** serving as the main power supply is your system's crucial first point of protection. It meticulously controls and safeguards the incoming AC power, & it's 230V single-phase or 415V three-phase.

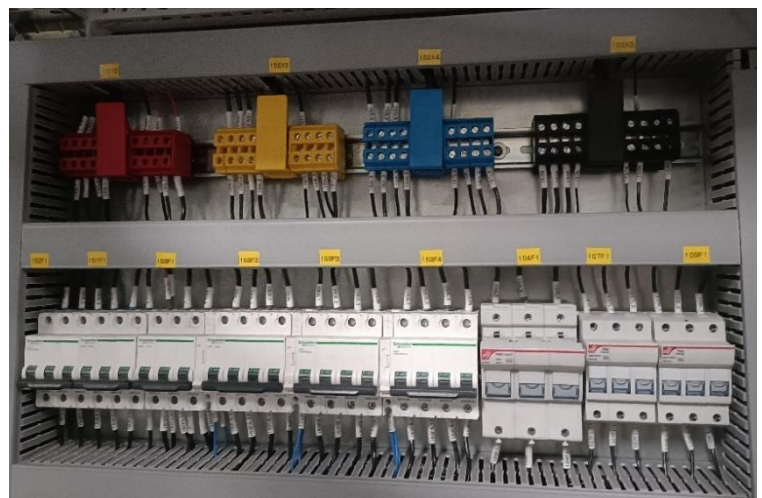
What the MCB Does for You:

- **Main ON/OFF Switch:** Functions as the primary power toggle for the entire machine.
- **Comprehensive Protection:** Shields your entire system from:
 - **Short circuits:** Prevents excessive current flow during faults.
 - **Overloads:** Trips when current exceeds safe limits.
 - **High current surges:** Protects against sudden, damaging spikes in current.
- **Power Distribution Control:** Manages the power supply to sensitive components like SMPS units, VFDs (Variable Frequency Drives), controllers, and sensors.



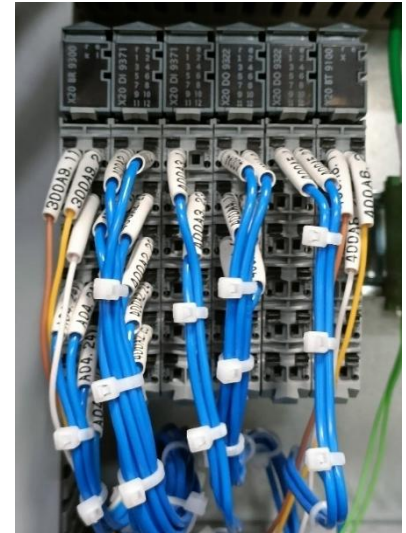
MCB TYPES WE USE –

- **SP (Single Pole):** Used for single-phase circuits like SMPS, lighting, and general sockets.
- **TP (Triple Pole):** Essential for safeguarding 3-phase main power inputs.
- **DP (Double Pole):** Ideal for single-phase applications requiring both Live and Neutral line cut-off.
- **MCB + RCCB Combo:** Provides enhanced safety and shock protection by combining overload/short-circuit protection with residual current detection.



- **PLC MODULE–**

In our Machine BB1500, the **PLC (Programmable Logic Controller) module** acts as the central brain, precisely automating critical operations such as tension control, braking, alignment, and managing various sensor inputs. The PLC efficiently processes information and issues commands:



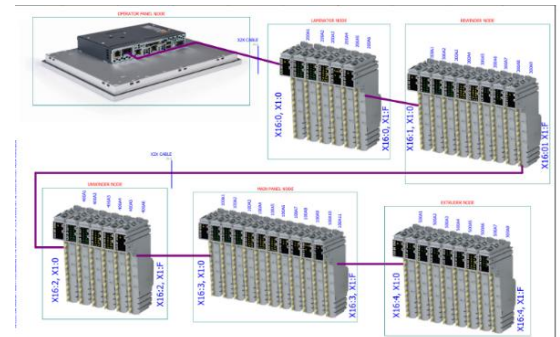
- **Receives Input From:**

- **Load cell / Tension sensor:** For continuous tension monitoring.
- **Proximity / Limit switches:** For detecting roll ends or machine positions.
- **Unwinder start/stop push buttons:** For operational commands.

- **Processes Logic:** Executes pre-programmed logic for precise control (e.g., determining optimal braking force, initiating stop sequences).

- **Sends Output To:**

- **Brake coil or clutch:** For tension and stopping control.
- **Motor starter:** For unwinder motor operation.
- **Alarms / Indicators:** For status alerts and warnings.
- **Web guide system:** For active alignment correction.



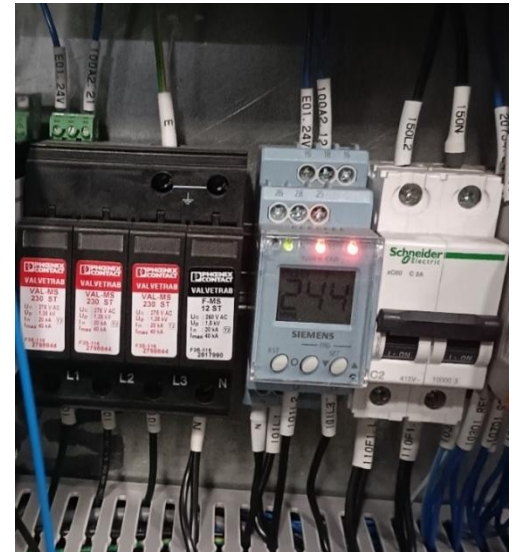
- **SPD & PHASE RELAY –**

A **Surge Protection Device (SPD)** is a critical safety component installed within our machine's control panel. Its primary function is to safeguard all sensitive electronic equipment from damaging high voltage spikes or surges.

The SPD acts as a rapid-response shield, specifically designed to:

- **Block Sudden Surges:** It immediately detects and blocks transient overvoltages caused by:
 - Lightning strikes
 - Power grid switching operations
 - Sudden motor start/stop events
 - **Divert Excess Energy:** Instead of allowing the surge to reach your valuable electronics, the SPD safely diverts this excess voltage directly to earth ground.
- This protects crucial components like:

- **SMPS (Switched Mode Power Supplies)**
- **PLCs (Programmable Logic Controllers)**
- **Controllers**
- **VFDs (Variable Frequency Drives)**



PHASE RELAY –

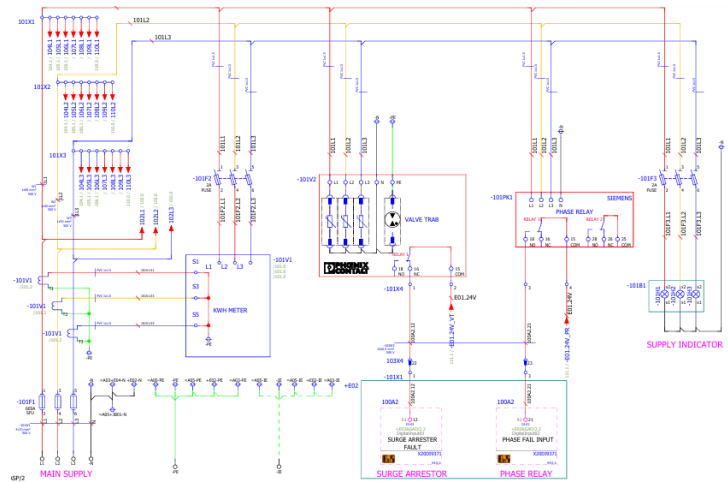
A **Phase Relay** is an indispensable monitoring device in our machines, specifically designed to supervise the incoming 3-phase power supply. Its essential role is to verify that the power is correct, balanced, and safe before allowing the machine to operate.

This intelligent relay constantly checks for critical power anomalies, including:

- **Phase Missing:** Detects if one or more phases are absent.
- **Wrong Phase Sequence:** Identifies incorrect rotation (e.g., R, Y, B phases reversed).
- **Phase Imbalance:** Monitors significant voltage differences between phases.
- **Overvoltage/Undervoltage:** Protects against excessively high or low supply voltages.

Where We Integrate the Phase Relay:

- **Just After the Main Power Supply:** Providing immediate monitoring at the point of entry.
- **Before the Main Contactor:** Acts as a safety interlock, preventing power flow to the machine if conditions are unsafe.
- **In all machines utilizing 3-phase motors or drives:** Essential for protecting phase-sensitive equipment.



Skills And Knowledge Gained

UNDERSTANDING ELECTRIC CIRCUITS –

Electrical and electronic systems, from the simplest gadgets to complex industrial machinery, are built upon a foundation of diverse components that control, conduct, convert, and protect electrical energy. To truly understand how circuits function, it's essential to grasp the critical roles played by these fundamental building blocks.

Here's an overview of some of the most important components you'll encounter and their essential uses:

1. Switches and Relays: The Circuit Controllers

At their core, **switches** are the fundamental control elements in any circuit, acting as gatekeepers for current flow.

Manual Control:

SPST (Single Pole Single Throw) & SPDT (Single Pole Double Throw) switches allow users to manually open a circuit (turn it off) or reroute current to a different path.

Toggle switches, push buttons (Normally Open (NO) / Normally Closed (NC)), and **DIP switches** provide tangible means for physical interaction, controlling everything from a simple light to intricate machinery operations.

Electrical Control:

Relays, such as the **SPST Relay**, are essentially electrically controlled switches. They are invaluable for switching high voltage or high current loads using only a low-power control signal, making them critical in automation, automotive systems, and essential protection circuits.



2. Resistors and Variable Resistors: Managing Current Flow

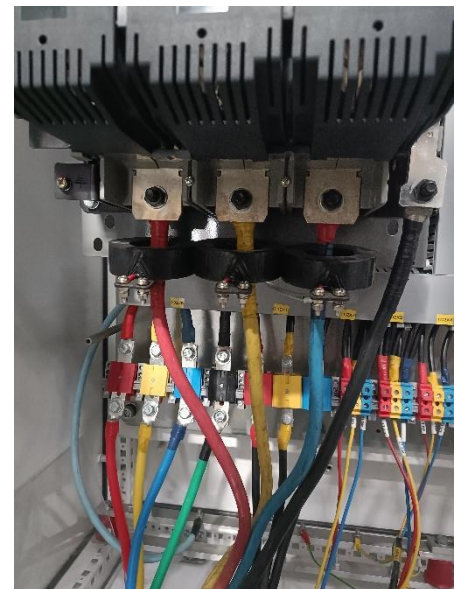
Resistors are passive components that oppose the flow of electric current, making them indispensable for:

- **Current Limiting:** Protecting sensitive components by reducing current to a safe level.
- **Voltage Division:** Creating specific voltage levels within a circuit.
- **Adjustable Resistance:**
 - **Variable resistors (rheostats)** and **potentiometers** allow for adjustable resistance, finding common use in applications like volume controls, light dimmers, and sensor calibration.
 - **Thermistors**, a specialized type of temperature-sensitive resistor, are crucial for precise temperature sensing and compensation circuits.

3. Power Sources: The Energy Providers –

Voltage sources and **current sources** form the very foundation of all electrical circuits, providing the energy needed for operation.

- **Direct Current (DC) Sources:** Such as **batteries**, provide a constant voltage or current flow in one direction.
- **Alternating Current (AC) Sources:** Typically from the **mains supply**, provide voltage or current that periodically reverses direction.
- **Specialized Sources: Controlled voltage/current sources** are used in advanced regulation circuits and for simulation purposes.
- **Energy Conversion:**



Generators, including **AC generators**, **pulse generators**, and **tachometer generators**, play a vital role in converting mechanical energy into electrical energy. They are fundamental to

renewable energy systems, automotive alternators, and for producing specific electrical signals for testing.

4. Motors and Actuators: Converting Electricity to Motion

Motors are dynamic components that convert electrical energy directly into mechanical motion, serving a vast array of applications:

- **Diverse Types: DC motors, AC motors, stepper motors, and linear motors** each offer unique characteristics suitable for specific applications—from simple cooling fans to complex robotic systems and industrial automation.
- **Essential Motor Parts:** Components like **carbon brushes** and different types of **windings** (series or shunt) are crucial internal elements that ensure efficient current flow and effective magnetic interaction within the motor.
- **Actuators:** These broader devices, including motors, convert electrical energy into physical action (e.g., opening a valve, moving a robotic arm).



5. Protection Devices: Safeguarding Circuits –

Protection devices are non-negotiable for the safety and longevity of any electrical system, shielding circuits from damage due to overload, short circuits, and faults.

- **Overcurrent Protection:**
 - **Fuses** and **circuit breakers** are designed to automatically interrupt current flow when it exceeds safe limits.
 - **Thermal fuses** offer protection against overheating.



Variants like **fast-blow** and **slow-blow fuses** are selected based on the specific response time required for the application.

- **Isolation: Isolator switches** manually disconnect a circuit for maintenance or safety.
- **Surge Protection: Lightning arresters** and **spark gaps** are vital for protecting large systems (like power grids and communication networks) from extreme voltage surges.

6. Sensors and Industrial Switches: Automation and Feedback –

These components are key to automation, providing critical feedback about a system's state or allowing manual control:

- **Automated Sensing:**
 - **Limit switches, proximity switches, float switches, and thermal switches** are used to detect conditions such as position, presence, liquid levels, or temperature, automatically triggering specific control actions.
- **Manual Control Interfaces:**
 - **Selector switches, joystick switches, and foot switches** provide operators with various intuitive methods for manual control in industrial machinery.



UNDERSTANDING PNEUMATIC CIRCUIT –

A pneumatic circuit is a system that uses compressed air to transmit and control energy. Understanding how to interpret and design these circuits is a valuable skill in many industrial and automation settings. Here Is the basic Pneumatic which is used in BB1500.

1. E2P Convertor –

The **E2P Converter**, or **Electrical to Pneumatic Converter**, is a crucial device in industrial automation, acting as a translator between electrical control signals and pneumatic actuation.

What it Does:

This device converts a standard **electrical current signal** (typically 4-20 mA) into a proportional **pneumatic pressure signal** (commonly 3-15 psi). Essentially, it allows an electrical control system to "talk" to a pneumatic device.

Where It's Used:

E2P converters are indispensable in **process industries** such as oil and gas, chemicals, and power generation. Their primary application is with **control valves**, where the main controller provides an electrical output, but the valve itself requires a pneumatic input to operate.

How It Works (Briefly):

1. **Electrical Input:** The converter receives an electrical current signal, usually within the 4-20 mA range.
2. **Internal Conversion:** Inside the converter, a precise mechanism—often a **force balance system** or a **nozzle-flapper system**—interprets this electrical signal.
3. **Pneumatic Output:** Based on the electrical input, the converter then produces a corresponding compressed air pressure output, typically ranging from 3-15 psi.

Its core purpose is to serve as a **bridge** between the electrical control system and the pneumatic actuator, ensuring seamless operation.

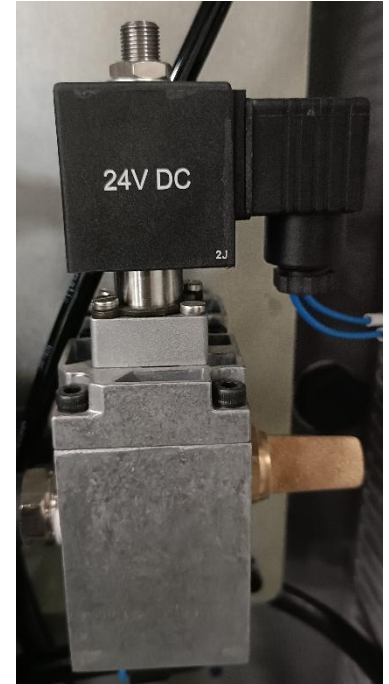


2. EXHAUST VALVE –

An **exhaust valve** is a critical component in any pneumatic system, primarily responsible for **safely releasing compressed air** from the system when pressure needs to be relieved or during a system shutdown. It's like the "relief valve" that ensures controlled depressurization.

Exhaust valves integral to various pneumatic applications:

- **Pneumatic Cylinders:** Essential for allowing cylinders to retract or extend efficiently by clearing air from the opposing chamber.
- **Control Valves:** Many directional control valves have integrated exhaust ports to release air when switching positions.
- **Safety Mechanisms:** They play a vital role in depressurizing entire pneumatic systems quickly in emergency situations or during maintenance, enhancing safety.



Types of Exhaust Valves:

Exhaust valves typically fall into two main categories:

- **Integrated Exhaust Port:** These are commonly found as built-in features within **directional control valves**, where air is vented directly from the valve's exhaust port.
- **External Exhaust Valve:** These are separate components added to a circuit specifically for **rapid air release**. A prime example is the **Quick Exhaust Valve (QEV)**.

3. FILTER, REGULATOR and LUBRICATION WITH DIGITAL GAUGE –

In pneumatic systems, the **FRL unit** is a critical 3-in-1 device designed for essential air preparation. Its primary goal is to enhance the **efficiency, reliability, and lifespan** of all downstream pneumatic components.

Filter: Ensuring Clean Air Supply

- **Function:** The filter's crucial role is to remove contaminants such as **dust particles, condensed water mist, oil aerosols, and other impurities** from the compressed air. It ensures that only clean, dry air enters the pneumatic circuit.
- **Why It's Needed:** Contaminated air is a major culprit for premature wear and tear. Dirty air can clog orifices, damage seals, corrode internal components of valves and actuators, leading to costly breakdowns and reduced system performance.

2. Regulator: Maintaining Stable Pressure

- **Function:** The regulator is responsible for **controlling and maintaining the compressed air pressure** at a precise, desired level (e.g., 5 bar, 70 psi). It takes the fluctuating incoming pressure from the compressor and delivers a stable, constant output pressure.
- **Why It's Needed:**
 - **Overpressure** can cause significant damage to pneumatic devices, exceeding their operational limits and potentially leading to safety hazards.
 - **Underpressure** results in insufficient force or erratic operation of actuators, leading to poor performance and inconsistent processes.



3. Lubricator: Extending Component Life

- **Function:** The lubricator introduces a **fine mist of lubricating oil** into the compressed air stream. This oil mist then travels through the system, coating the internal moving parts of pneumatic tools, cylinders, and certain valves.
- **Why It's Needed:** Proper lubrication is vital for:
 - Ensuring **smooth and friction-free movement** of pistons and other components.

- **Reducing wear and tear**, thereby significantly extending the service life of expensive pneumatic devices.
- Preventing rust and corrosion in susceptible parts.

4. Digital Pressure Gauge

While not strictly part of the FRL triplet, a **Digital Pressure Gauge** is an increasingly common and valuable addition to pneumatic systems, often integrated near the FRL or at critical pressure points.

- **Function:** Unlike traditional analog gauges, a digital pressure gauge provides a **real-time, highly accurate digital display** of the air pressure (in units like bar, psi, kPa, etc.). It replaces the need for manual interpretation of needle positions.
- **Advantages:**
 - **High Visibility:** Clear, easy-to-read digital readouts.
 - **Precise Reading:** Offers superior accuracy compared to analog counterparts, reducing human error.
 - **Enhanced Features:** Many digital gauges come with advanced functionalities such as **alarm outputs** (to signal over or under pressure), **data logging capabilities**, and **communication interfaces** for integration with control systems.

5. THE PNEUMATIC MANIFOLD –

A **Pneumatic Manifold**, often referred to as a **Mainfold Block** or **Distributor Block**, is a fundamental component in pneumatic systems. Its primary function is to efficiently **take a single input of compressed air and distribute it into multiple separate output lines**, feeding various pneumatic devices.

The manifold acts as a central hub for air distribution:

- **Single Source Input:** It receives compressed air from a single supply line, typically originating from the compressor or the FRL (Filter-Regulator-Lubricator) unit.
- **Multi-Point Distribution:** From this single input, the manifold branches out to provide individual air supplies to numerous pneumatic components, such as cylinders, control valves, pneumatic tools, and other actuators.

How It Works:

1. **Input Port (Inlet):** Compressed air enters the manifold through a single inlet port, which is connected to the main air supply.
2. **Internal Channeling:** Inside the manifold, precision-engineered internal channels or passages direct the incoming air, ensuring it is distributed evenly to all the output ports.
3. **Multiple Output Ports (Outlets):** Each of these outlet ports serves as a dedicated connection point for an individual pneumatic device, allowing for independent air supply to each component.



Why Pneumatic Manifolds Are Indispensable:

Manifolds offer significant advantages in pneumatic system design and operation:

- **Compact and Clean Layout:** They eliminate the need for extensive individual tubing runs from the main supply to each component, resulting in a much neater, more organized, and space-saving system layout.
- **Centralized Air Distribution:** All connections for a specific area or function can be consolidated at a single point, simplifying system design and troubleshooting.
- **Simplified Maintenance:** With centralized distribution, identifying and isolating issues, or performing maintenance on individual lines, becomes much easier.

- **Time and Cost Savings:** In systems with numerous pneumatic devices, manifolds significantly reduce installation time and material costs associated with tubing, fittings, and connectors.

These are the base of Pneumatic if any Output is not coming properly It can affect the whole system. All valve connections are Interconnected.

6. Pressure Gauge –

A **Pressure Gauge** is an essential instrument in any pneumatic system, designed to **measure and visually display the pressure** of compressed air within pipelines, cylinders, valves, and other components. It provides critical real-time feedback on system conditions.

Types of Pressure Gauges:

- **Analog Gauge:** This is the traditional type, featuring a **needle and a calibrated dial** (e.g., 0-10 bar). They are simple, robust, and commonly used for general pressure indication.
- **Digital Gauge:** These modern gauges utilize an **LCD or LED display** to show pressure readings. They offer **higher precision, better readability**, and often come with advanced features.

Key Functions:

- **Real-time Monitoring:** Provides an immediate visual indication of pressure at various points within the pneumatic circuit.
- **Preventing Damage:** Helps in identifying and preventing conditions of **overpressure**, which can severely damage components and pose safety risks.
- **Ensuring Optimal Performance:** Allows operators to verify that the system is running at its **correct and specified operating pressure**, crucial for consistent performance of actuators and tools.

Units Displayed:

Pressure gauges can display readings in various standard units, including:

- **bar**
- **psi** (pounds per square inch)
- **MPa** (megapascals)
- **kPa** (kilopascals)

Strategic Placement:

For effective monitoring, pressure gauges are strategically placed throughout a pneumatic system, commonly found:

- Near **air compressors** and **FRL (Filter-Regulator-Lubricator)** units.
- After **pressure regulators** to confirm output pressure.
- On **pneumatic cylinders** to monitor chamber pressures.
- At critical points within **valve manifolds** or branch lines.



The Pressure Controller: Automated Pressure Management

A **Pressure Controller** is an advanced device that goes beyond mere measurement; it actively **monitors and automatically adjusts pressure** within a pneumatic system to maintain a desired setpoint. It's the "brain" for pressure regulation.

Core Functionality:

The controller operates on a feedback loop principle:

1. **Pressure Input:** It receives real-time pressure readings from connected **sensors or pressure gauges**.
2. **Comparison:** It compares the measured pressure against a pre-programmed **set value** (the desired pressure).
3. **Output Adjustment:** Based on this comparison, it sends a corrective output signal to **control valves, proportional regulators, or even the compressor** to increase or decrease pressure as needed, thereby maintaining the setpoint.

Types of Pressure Controllers:

- **Manual Controllers:** These are basic devices where the pressure setpoint is adjusted by hand using **knobs or screws**. They offer simple, direct control.
- **Electronic Controllers:** More sophisticated, these utilize **sensors combined with logic circuits** (e.g., **PID controllers**) to provide precise and dynamic pressure regulation.
- **Smart Controllers:** These are modern, often network-enabled devices featuring **digital interfaces, programmable logic**, and advanced communication capabilities for complex control strategies and integration into larger automation systems.

Basic Working Principle:

A **pressure sensor** continuously reads the current pressure in the system.

This reading is fed to the **controller**, which compares it to the predefined desired **setpoint**.

If the measured pressure deviates from the setpoint (either too high or too low), the controller calculates the necessary correction and sends an appropriate signal to an **actuator** (like a proportional valve or a compressor control), which then adjusts the pressure back to the desired level.

7. SOLENOID VALVE 3/2 –

A **3/2 Solenoid Valve**, also known as a **Three-Port Two-Position Valve**, is a fundamental component in pneumatic systems that uses an electrical signal to control the flow of compressed air. It's particularly well-suited for controlling single-acting cylinders.

Decoding "3/2":

The "3/2" nomenclature clearly defines the valve's configuration:

- **3 Ports:**
 - **P (Pressure Inlet):** This is where compressed air from the supply (e.g., compressor or FRL unit) enters the valve.
 - **A (Outlet):** This port directs the controlled air flow to the pneumatic actuator (e.g., a single-acting cylinder or a specific part of a larger pneumatic circuit).
 - **R (Exhaust):** This port is used to safely release "spent" or excess air from the actuator or the valve itself to the atmosphere when the valve is in its de-energized state.
- **2 Positions:**
 - **Energized (Coil ON):** This is the active state when electrical current flows through the solenoid coil.
 - **De-energized (Coil OFF):** This is the passive or default state when no electrical current flows through the solenoid coil.

Working Principle: Electrically Controlled Air Flow

The valve switches the flow path of compressed air based on the electrical state of its solenoid coil.

Solenoid Coil State	Action	Air Flow Path
OFF (De-energized)	Default/Rest Position	P blocked or A → R (Exhaust)
ON (Energized)	Activated Position	P → A (Air supplied to actuator)

How It Works (Simple Explanation):

- **When OFF (De-energized):**
 - In its default state, the valve typically blocks the incoming air supply (P) from reaching the outlet (A).
 - Crucially, it also connects the outlet (A) to the exhaust (R), allowing any air present in the connected actuator to be released. This action causes a single acting cylinder, for example, to retract to its original position, often assisted by a spring.
- **When ON (Energized):**
 - When an electrical current is applied to the solenoid coil, it creates an electromagnetic field that shifts the internal mechanism of the valve.
 - This action opens the path from the pressure inlet (P) to the outlet (A), allowing compressed air to flow to the actuator.

Common Applications:

3/2 solenoid valves are widely used for:

- **Control of Single-Acting Cylinders:** They are the standard choice for extending and retracting single-acting pneumatic cylinders.
- **Start/Stop Air Flow in Automation:** Used to initiate or halt air flow to specific parts of a pneumatic circuit based on electrical signals from a control system (e.g., PLC).
- **Emergency Shut-off Systems:** Can be employed as safety valves to rapidly block or vent air in critical situations.
- **Pilot Control:** Serving as pilot valves to actuate larger, pneumatically controlled valves.



DEVICES USED THROUGH OUT THE INTERNSHIP PERIOD

Multimeter –

A **multimeter** is an essential electronic measuring instrument used in labs, industries, and field work. It combines multiple functions in a single device, typically including the ability to measure **voltage**, **current**, and **resistance**. Modern digital multimeters (DMMs) may also include **continuity testing**, **diode testing**, **capacitance**, **frequency**, **temperature**, and more.

There are two types:

- **Analog Multimeter** – uses a moving needle.
- **Digital Multimeter (DMM)** – shows readings on a digital display.

Main Modes of a Multimeter –

1. DC Voltage (V— or VDC)

- **Use:** To measure voltage from DC sources like batteries, power supplies, and circuits.
- **Symbol:** V— or VDC
- **How to use:** Connect the red probe to the positive terminal and the black probe to the ground or negative side.
- **Common range:** 200 mV to 1000 V

2. AC Voltage (V~ or VAC)

- **Use:** To measure voltage in AC systems like wall outlets.
- **Symbol:** V~ or VAC
- **How to use:** Connect the probes across the live and neutral wires.
- **Typical range:** Up to 750 V or 1000 V





3. DC Current (A— or ADC)

- **Use:** To measure current in DC circuits like motor drives, solar systems.
- **Symbol:** A— or ADC
- **How to use:** Break the circuit and connect the multimeter **in series**.
- **Note:** Always check the correct port (10A/ μ A/mA) before measuring.

4. AC Current (A~ or AAC)

- **Use:** To measure current in AC circuits.
- **Symbol:** A~ or AAC
- **How to use:** Similar to DC current — connect in series with the load.

5. Resistance (Ω)

- **Use:** To measure the resistance of resistors, wires, or coils.
- **Symbol:** Ω
- **How to use:** Connect probes across the component (circuit should be OFF).
- **Range:** Ohms (Ω), kilo-ohms ($k\Omega$), mega-ohms ($M\Omega$)

6. Continuity Test (or diode with sound icon)

- **Use:** To check if a wire or connection is complete.
- **Symbol:** Sound wave or diode symbol
- **How to use:** Connect probes; if there is a short path (low resistance), the meter **beeps**.

7. Diode Test Mode (diode symbol)

- **Use:** To check if a diode is working.
- **Symbol:** Diode arrow
- **How to use:** Forward bias gives a reading (0.5–0.8V); reverse bias gives OL (open line).

8. Capacitance Mode (F or two parallel lines)

- **Use:** To measure the capacitance of capacitors.
- **Symbol:** F or ||
- **How to use:** Discharge the capacitor first, then measure across its terminals.

9. Frequency (Hz)

- **Use:** To measure the frequency of AC signals or PWM signals.
- **Symbol:** Hz
- **How to use:** Connect across the waveform source.

10. Temperature (°C/°F)

- **Use:** To measure temperature using a thermocouple probe.
- **Symbol:** °C or °F
- **How to use:** Plug in the K-type thermocouple and place it where needed.

11. Transistor Testing (hFE)

- **Use:** To test transistor gain (hFE value).
- **Symbol:** hFE
- **How to use:** Insert the transistor legs into the specific socket marked E-B-C.

12. Auto Range / Manual Range

- **Use:** Some meters allow you to choose ranges manually (e.g., 200V, 600V) or automatically detect.
- **Auto Range:** Easier and faster for beginners.
- **Manual Range:** Gives more control, useful for sensitive components.

Important Safety Tips

- Always start on the highest range to avoid damage.

- Never measure resistance or continuity in a powered circuit.
- Use appropriate ports (COM, V Ω mA, 10A) depending on measurement.
- Keep hands behind the safety line when using probes.

The Tachometer: Measuring Rotational Speed in Automation

A **Tachometer** is an essential instrument engineered to measure the **rotational speed** of a shaft, motor, or any rotating mechanical component. Its primary output is typically displayed in **Revolutions Per Minute (RPM)**, making it a critical tool across diverse sectors such as automotive, industrial manufacturing, and advanced automation systems.

Types of Tachometers –

Tachometers primarily fall into two broad categories based on their display and operating principles:

a) Analog Tachometer

- **Display:** Features a traditional needle dial.
- **Operation:** Relies on either mechanical linkages or electromagnetic induction to measure and display speed.
- **Common Use:** Historically prevalent in vehicle dashboards for engine RPM monitoring.

b) Digital Tachometer

- **Display:** Presents RPM values numerically on a digital screen.
- **Operation:** Utilizes advanced sensors coupled with microcontrollers to achieve highly accurate readings.



- **Preference:** The go-to choice in modern industrial and automation systems due to its precision and ease of integration.

Working Principle: How Rotational Speed is Detected

Tachometers operate by detecting rotational motion and converting it into a quantifiable electrical signal. The specific method of detection varies by type:

- **Optical Sensors:** These non-contact sensors detect light reflections from, or interruptions by, a rotating target (e.g., a piece of reflective tape or a slotted disk). Each reflection or interruption corresponds to a rotation or a segment of a rotation.
- **Magnetic Sensors:** These inductive or Hall-effect sensors detect changes in a magnetic field caused by the passage of ferrous components like gear teeth or integrated magnets on the rotating shaft.
- **Proximity Sensors:** Operating on a non-contact basis, these sensors detect the presence or absence of metal parts as they pass a fixed point on the rotating assembly.

In all these cases, the sensors generate **pulses** for each revolution or a defined segment of a revolution. These pulses are then processed by the tachometer's internal circuitry to precisely calculate and display the RPM.

Operating a Digital Tachometer: A Step-by-Step Guide

Using a digital tachometer, especially the non-contact type, is generally straightforward:

1. **Prepare the Target:** Apply a piece of **reflective tape** or ensure a suitable contrasting surface on the rotating shaft or component.
2. **Aim the Sensor:** Point the tachometer's integrated **laser or infrared (IR) sensor** directly at the prepared tape area.
3. **Activate the Device:** Turn on the tachometer. It will immediately begin detecting reflections or signals and counting the corresponding pulses.

4. **Read the RPM:** The calculated rotational speed will be displayed numerically on the digital screen.

Diverse Applications of Tachometers

The versatility of tachometers makes them indispensable across numerous industries:

- **Automotive Industry:** Crucial for monitoring **engine speed (RPM)** in vehicles, aiding in gear shifting and performance optimization.
- **Industrial Machinery:** Used to verify and maintain the correct operating speeds of **motors, fans, pumps, and conveyor belts**.
- **Laboratories and R&D:** Essential for **testing and analyzing** the performance of rotating systems and prototypes.
- **Automation Systems:** Provide vital **feedback** in closed-loop motor control systems, enabling precise speed regulation and positional accuracy.

Advantages of Using Tachometers

- **Real-time Monitoring:** Offers instantaneous display of RPM, allowing for immediate adjustments.
- **Speed Regulation & Diagnostics:** Facilitates precise control of motor speeds and assists in diagnosing operational issues.
- **Non-Contact Operation:** Many modern tachometers are non-contact, minimizing wear and tear on the measured component and ensuring safety.

Limitations to Consider

Despite their advantages, tachometers do have certain limitations:

- **Line of Sight:** Optical tachometers require a clear, unobstructed line of sight to the rotating target.
- **Environmental Sensitivity:** Performance can be affected by environmental factors such as excessive **dust** or **vibration**.

- **Calibration Needs:** For applications demanding extremely high precision, periodic **calibration** of the tachometer may be necessary.

Clamp Meter –

A **Clamp Meter** is an indispensable handheld electrical testing tool primarily designed to measure **Alternating Current (AC)** and, in more advanced models, **Direct Current (DC)**, *without the need to break the circuit*. It ingeniously combines the core functionalities of a digital multimeter (DMM) with a unique current sensing clamp. This clamp can be safely and easily secured around an electrical conductor to measure current flow, making it a highly efficient and secure diagnostic device.

Key Feature: The Current-Sensing Clamp

The defining characteristic of a clamp meter lies in its **clamp-shaped jaws**. These jaws are engineered to open and close around an electrical conductor. The innovative design leverages the **magnetic field** generated by current flowing through a wire to accurately measure the magnitude of that current. This principle is based on **Faraday's Law of Electromagnetic Induction**, allowing for contactless and safe current measurement.

Versatile Modes and Measurements

Modern clamp meters are highly versatile, offering a range of measurement modes beyond just current:



- **AC Current (A~)**
 - Measures Alternating Current without requiring direct electrical contact with the conductor.
 - Commonly has a wide measurement range, often up to 1000 Amperes.
 - **Crucial:** Must be clamped around a *single* conductor for accurate readings.
- **DC Current (A=)** (Available in advanced models)
 - Measures Direct Current by utilizing a **Hall-effect sensor** within the clamp.
 - Essential for applications involving batteries, solar power systems, and DC motor circuits.
- **Voltage (V~ and V=)**
 - Measures both AC and DC voltage, functioning identically to a standard multimeter when using its test probes.
- **Resistance (Ω)**
 - Measures the electrical resistance of components like resistors, wiring, or motor windings.
 - **Important:** The circuit being tested must be turned off and de-energized.
- **Frequency (Hz)**
 - Measures the frequency of AC signals, useful for analyzing power quality or motor speed.

How to Use a Clamp Meter (for Current Measurement)

Measuring current with a clamp meter is a simple, straightforward process:

1. **Select Mode:** Turn on the clamp meter and set the rotary dial to the appropriate **AC current (A~)** or **DC current (A=)** mode.
2. **Isolate Conductor:** Open the clamp jaws and carefully place them around **one single wire**. It is critical *not* to clamp around both the live and neutral (or positive and negative) wires of a circuit, as their magnetic fields will cancel each other out, resulting in a zero reading.

3. **Read Value:** The current value will be displayed on the meter's digital screen.
4. **Other Measurements:** For voltage, resistance, or continuity tests, use the provided test leads just as you would with a conventional digital multimeter.

Advantages of Using a Clamp Meter

The unique design of a clamp meter offers significant benefits for electrical testing:

- **Non-Intrusive:** Eliminates the need to disconnect or break the circuit, saving time and reducing potential hazards.
- **Enhanced Safety:** Allows for current measurement without direct physical contact with live wires, significantly improving user safety.
- **Speed and Convenience:** Provides instantaneous current readings simply by clamping around a conductor.
- **Multifunctional:** Beyond current, it also measures voltage, resistance, continuity, and often other parameters, making it a versatile diagnostic tool.

Common Applications

Clamp meters are essential tools across a wide range of industries and scenarios:

- **Industrial Maintenance:** Ideal for troubleshooting motors, control panels, heavy current lines, and industrial machinery.
- **HVAC and Electrical Troubleshooting:** Used by technicians for diagnostics in heating, ventilation, air conditioning, and general electrical systems.
- **Solar and Battery Systems:** Crucial for measuring DC currents in photovoltaic arrays and battery banks.
- **Electricians and Field Engineers:** A go-to tool for on-site diagnostics, installation checks, and verifying electrical loads.
- **Home Installations and Repair:** Useful for household electrical checks, appliance repair, and verifying circuit loads.

Essential Safety Precautions –

When using a clamp meter, always adhere to these safety guidelines to ensure accurate readings and prevent injury:

- **Single Conductor Rule:** Always clamp around **only one conductor** to obtain a correct current reading.
- **Correct Setting:** Ensure the meter is set to the **appropriate AC or DC current mode** for the circuit being tested.
- **Cable Types:** Avoid clamping on multi-core cables or heavily sheathed wires unless you can safely isolate a single conductor.
- **Rated Limits:** **Never exceed the meter's rated input limits** for voltage or current, which are typically marked on the device.

CLAMP AMMETER –

A **Clamp Ammeter**, also commonly referred to as a Clamp Current Meter, is a specialized handheld electrical measuring instrument designed to measure electric current in a conductor **without requiring direct contact or breaking the electrical circuit**. The device operates by physically clamping its jaws around a single current-carrying wire, detecting the magnetic field produced by the current flow, and then converting this magnetic field into a quantifiable current value, displayed in Amperes (A).

2. Working Principle

The fundamental operation of a clamp ammeter is rooted in **Faraday's Law of Electromagnetic Induction**:

When an electric current flows through a conductor, it generates a proportional magnetic field around that conductor. The clamp's jaws, which typically contain a magnetic core and a sensor, detect this induced magnetic field. This detected magnetic field is then precisely

translated into a proportional electrical signal, which the meter's internal circuitry processes and displays as the current magnitude in amperes.

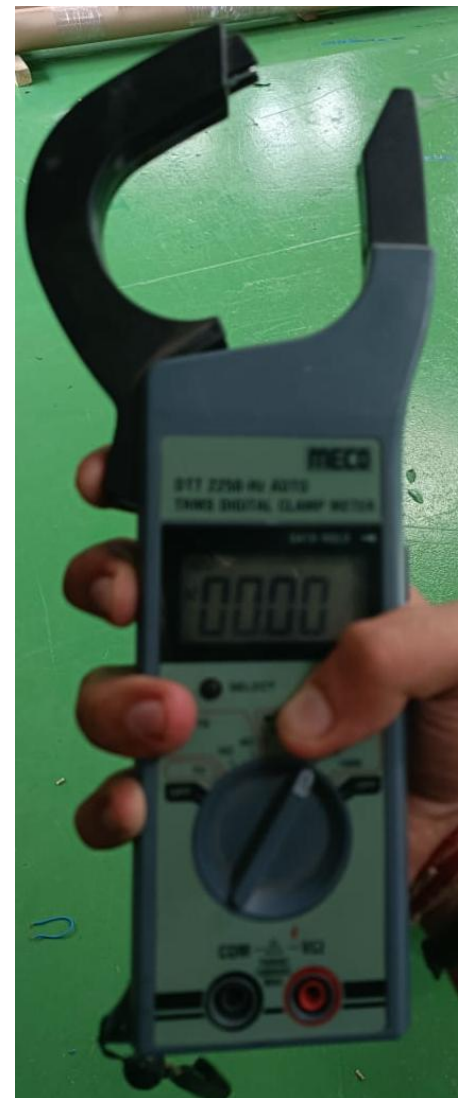
Modern digital clamp ammeters, especially those capable of measuring both AC and DC currents, commonly employ a **Hall-effect sensor** within the clamp to achieve this non-contact measurement.

3. How to Use a Clamp Ammeter (for Current Measurement)

Using a clamp ammeter for current measurement is a straightforward process:

1. **Power On and Select Mode:** Turn on the clamp ammeter and set its rotary dial to the appropriate current measurement range (AC current "A~" or DC current "A=").
2. **Open and Clamp:** Use the trigger mechanism to open the clamp jaws and carefully position them around a **single, isolated live conductor**.
3. **Center the Wire:** Ensure the wire is as centrally located as possible within the clamp jaws for the most accurate reading.
4. **Read Display:** The measured current value will be displayed directly on the meter's digital screen.

Crucial Note: It is imperative **never to clamp around an entire multi-core cable** (e.g., a cable containing both live and neutral wires, or positive and negative wires together). Doing so will result in the magnetic fields from opposing currents canceling each other out, leading to an erroneous zero current reading.



4. Common Modes Available in Clamp Ammeters

While dedicated clamp ammeters focus solely on current, many modern devices offer multi-functional capabilities:

Mode	Symbol	Description
AC Current	A~	Measures alternating current, common in motors, panels, and HVAC systems.
DC Current	A=	Measures direct current, crucial for batteries, solar panels, and electric vehicles.
Voltage	V~, V=	Measures both AC and DC voltage using external test probes (if supported).
Continuity		Checks for a closed circuit by emitting an audible beep when electrical continuity is detected.
Resistance	Ω	Measures the electrical resistance of components or circuits. (Requires circuit to be de-energized)
Frequency	Hz	Measures the frequency of AC signals.
Capacitance	-	Measures capacitance (in advanced models).
Temperature	$^{\circ}\text{C}/^{\circ}\text{F}$	Measures temperature using a K-type thermocouple (in advanced models).

5. Key Applications of Clamp Ammeters

The utility of clamp ammeters spans numerous industries and diagnostic tasks:

- **Electrical Panel Troubleshooting:** Quickly identify overloaded circuits or unbalanced loads.
- **Industrial Machine Monitoring:** Assess current draw of motors, pumps, and other machinery to detect anomalies.
- **Motor Maintenance:** Verify operating currents to ensure efficiency and prevent overheating.

- **Battery Bank Current Checks:** Monitor charge/discharge currents in battery systems.
- **Solar Panel System Testing:** Measure current output from solar arrays and charge controllers.
- **Electric Vehicle (EV) Current Monitoring:** Diagnose current flow in EV charging and power systems.
- **Safety Checks:** Perform live circuit measurements without the risk of breaking connections or direct contact.

6. Advantages

The clamp ammeter offers several significant advantages over traditional inline current measurement methods:

- **Non-Intrusive:** Eliminates the need to open or break electrical circuits, saving time and reducing potential hazards.
- **Quick & Convenient:** Enables rapid current measurements simply by clamping around a wire.
- **Enhanced Safety:** Reduces the risk of electrical shock by removing the need for direct electrical contact with live conductors during current measurement.
- **Handles High Current:** Many models are designed to measure very high currents, sometimes up to 2000 Amperes or more, which would be impractical for standard multimeters.
- **Portable:** Typically lightweight, handheld, and battery-operated, making them ideal for field use.

7. Safety Tips

Adhering to safety protocols is paramount when using any electrical testing equipment:

- **One Conductor Rule:** Always clamp around **only one conductor** to ensure accurate readings and avoid magnetic field cancellation.

- **Insulation Integrity:** Never clamp around wet or visibly damaged wire insulation.
- **Range Verification:** Always check the **maximum current and voltage rating** of the clamp ammeter before use, ensuring it is suitable for the circuit being tested.
- **Personal Protective Equipment (PPE):** For high-voltage or high-current applications, always use appropriate PPE, including insulated gloves and safety glasses.
- **Resolution for Low Current:** Avoid using standard clamp ammeters for very low current (milliampere) applications unless the specific model explicitly supports mA resolution, as accuracy will be poor.

8. Limitations

Despite their versatility, clamp ammeters do have some limitations:

- **Low Current Sensitivity:** Not ideally suited for measuring very low currents (e.g., in the milliampere range) due to lower sensitivity compared to inline measurements.
- **Accuracy:** While generally good, their accuracy may be slightly lower than direct inline measurement methods (e.g., shunts) in some high-precision applications.
- **Limited Measurement Scope:** Cannot directly measure electrical energy (kWh) or power (Watts); they only measure current.
- **Physical Clearance:** Requires sufficient physical clearance around the wire for the jaws to properly clamp.

9. Conclusion

The **Clamp Ammeter** is an indispensable and highly valuable tool in the toolkit of any electrician, maintenance technician, or field engineer. Its ability to perform safe, fast, and non-invasive current measurements, particularly in live circuits, makes it crucial for diagnostics, troubleshooting, and routine electrical maintenance. A thorough understanding of its working principle, correct usage procedures, and associated safety precautions is essential for efficient and safe electrical diagnostics.

DRIVE PARAMETER

Here is the list of the drive Which we use in BB1500 machine –

1. Extruder Drive –

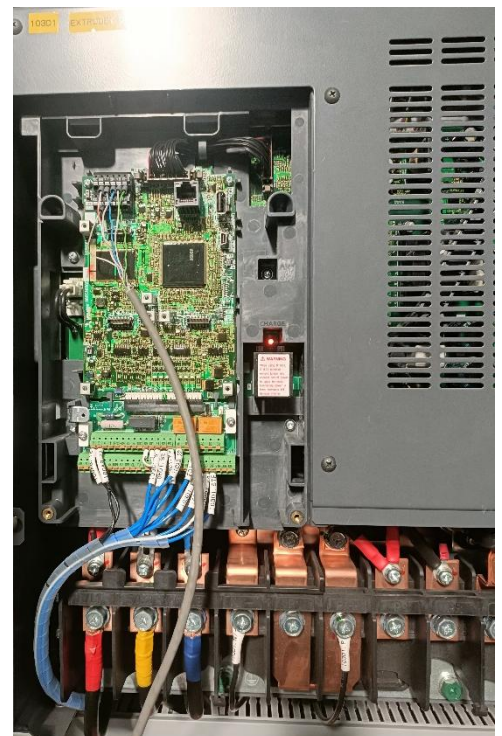
An **Extruder Drive** refers to a sophisticated motor and drive system specifically engineered to **control the speed and torque of the screw within BB1500 machine.**

The primary role of the extruder drive is to ensure the **precise and consistent rotation of the extruder screw.** This screw is vital for:

1. **Material Conveyance:** Pushing raw material (e.g., plastic pellets, rubber compounds, food dough) forward.
2. **Melting and Mixing:** Generating shear heat and uniformly mixing the material as it moves through heated barrel zones.
3. **Forcing Through the T SHAPED Die:** Building the necessary pressure to force the molten or processed material through a die, forming continuous shapes such as pipes, sheets, films, profiles, or even complex food products.

Extrusion processes demand precise and robust control, necessitating a dedicated drive system due to several factors:

- **Precise Speed Control:** Critical for maintaining a uniform material flow rate, which directly impacts product quality, consistency, and dimensions.
- **High Torque at Low Speed:** Essential during the initial startup phase when the screw is loaded with cold, viscous material, and throughout processing of heavy or high-viscosity materials.



- **Energy Efficiency:** The ability to vary screw speed based on material properties, product specifications, and production rates allows for significant energy savings compared to fixed-speed operations.
- **Reverse Functionality:** Often required for clearing material jams, facilitating cleaning procedures, or reversing the screw during specific process steps.

2. Laminator Drive-

A **Laminator Drive** constitutes the integrated motor and drive system specifically designed to **control the precise rotational speed of the rollers within a lamination machine**. These machines are fundamental in various industries for bonding two or more layers of diverse materials—such as plastic films, paper, BOPP (Biaxially Oriented Polypropylene), and adhesives—under controlled conditions of heat and pressure.

Main Purpose of the Drive in a Laminator:

The drive system is pivotal for achieving high-quality lamination and ensuring efficient operation:

- **Precise Roller Speed:** Maintaining consistent and accurate roller speed is paramount for uniform material feeding and product quality.
- **Tension Control:** It is crucial for managing the tension of the film or web materials between unwinding and winding stations, preventing stretching, sagging, or tearing.
- **Synchronization:** The laminator drive must synchronize seamlessly with other drives in the line, including unwinder, extruder (if integrated), and rewinder drives, to ensure continuous and harmonious material flow.
- **Defect Prevention:** Precise control helps to eliminate common lamination defects such as wrinkles, air bubbles, and material misalignment.

Why a Dedicated Drive System is Essential:

Lamination processes demand sophisticated control due to the sensitive nature of the materials and the need for flawless output:

- **Variable Speed Control:** Different materials, thicknesses, and adhesive types require specific processing speeds to achieve optimal bonding. The drive must offer fine-tuned speed adjustment.
- **Smooth Start/Stop:** Abrupt changes in speed can cause sudden jerks in web tension, leading to material breakage or damage. A smooth acceleration/deceleration profile is vital.
- **Tension Control Integration:** The drive system often integrates with tension control mechanisms to keep the web material uniformly stretched throughout the lamination process, which is crucial for preventing dimensional instability and defects.
- **Speed Synchronization:** In multi-stage lamination lines, the drive must accurately synchronize its speed with upstream (e.g., extruder, unwinder) and downstream (e.g., nip roll, rewinder) sections to prevent material buildup or excessive tension.

Typical Drive Technologies Employed:

The selection of drive technology depends on the desired level of precision, dynamic response, and cost-effectiveness:

- **AC VFD (Variable Frequency Drive):** These are the most commonly used drives, controlling standard **3-phase AC induction motors**. They provide excellent speed control, energy efficiency, and are robust enough for most lamination applications.
- **Closed-loop Vector Drive:** For applications requiring high accuracy in speed and torque, especially where tension control is critical, vector control VFDs are preferred. They use feedback (e.g., from an encoder) to precisely control motor flux and torque.
- **Servo Drives:** Utilized in high-end lamination machines where **extreme precision, high-speed responsiveness, and dynamic accuracy** are paramount. Servo systems offer superior performance for demanding applications.



3. Melt Pump Drive-

A **Melt Pump Drive** is a specialized motor and drive system specifically engineered to control the motor that powers a **melt pump**. The melt pump itself is a crucial component in polymer processing lines (such as film, sheet, fiber, and profile extrusion), designed to **deliver molten polymer or plastic at a precise, constant pressure and flow rate** from the extruder to the downstream die.

Core Function:

The melt pump drive system plays a critical role in refining the output from the extruder:

- **Pressure and Flow Stabilization:** Even if the extruder's output exhibits fluctuations in flow or pressure (which is common due to the nature of screw extrusion), the melt pump, controlled by its dedicated drive, ensures a **steady, non-pulsating, and precisely controlled output** to the die.
- **Uniform Product Quality:** This stable delivery is absolutely critical for achieving **uniform product thickness, consistent dimensions, and overall high quality** in the final extruded product.



Why a Melt Pump Requires a Dedicated Drive:

The necessity for a separate, dedicated drive for the melt pump stems from several key process requirements:

- **Precise Flow Rate Control:** It allows for accurate control over the volumetric flow rate (often expressed in kg/h or l/min) of the molten polymer, directly impacting the final product's gauge and output.

- **Maintaining Steady Output Pressure:** The drive enables the melt pump to actively maintain a constant pressure at the die, isolating the die from pressure variations originating from the extruder.
- **Pulsation Dampening:** The melt pump significantly reduces or eliminates pressure pulsations inherent in the extruder's screw rotation, leading to a much smoother and more consistent melt stream.
- **Protection of Downstream Equipment:** By stabilizing pressure and flow, the melt pump protects sensitive downstream components like screen changers, static mixers, and expensive dies from harmful pressure spikes or fluctuations.

Typical Drive Technologies Employed:

Given the demands for high precision and stability, specific types of drives are preferred for melt pump applications:

- **AC Vector VFD (Closed-Loop):** This is the most common and highly effective solution. **Closed-loop vector control** (using an encoder for precise speed feedback) allows the drive to achieve extremely accurate speed and torque control over the motor, which is crucial for maintaining constant flow and pressure.
- **Servo Drives (Optional):** In applications demanding the **highest levels of accuracy, dynamic response, and very fine control** over melt delivery, servo drives may be utilized. They offer superior performance in terms of speed holding and torque response.
- **Feedback Integration:** Melt pump drives are almost always paired with a **pressure sensor** (located before or after the pump) or a **load cell** (measuring motor torque) as feedback. This allows the drive to operate in a closed-loop control mode, continuously adjusting pump speed to maintain the desired pressure or flow rate.

4. Auger Motor Drive

An **Auger Motor Drive** refers to the integrated motor and drive system responsible for controlling the rotation of an **auger screw** (a helical screw conveyor). These systems are ubiquitous across various industries, including plastics, food processing, pharmaceuticals, and packaging, where they are used to accurately move and dispense solid, semi-solid, or powdered materials.

Core Function:

The auger screw, typically housed within a tube or trough, rotates to **feed or precisely dose** materials. The drive system's primary role is to control the **speed and torque** of this rotation, directly influencing the material's flow rate and accuracy.

Why a Dedicated Drive System is Essential for Auger Motors:

The unique demands of material handling via augers necessitate a specialized drive system for several reasons:

- **Variable Speed Control:** Essential for precisely adjusting the material feed rate (e.g., in kg/h or g/shot) to meet production requirements and material specifications.
- **High Torque at Low Speed:** Crucial for effectively handling heavy, dense, or sticky materials, especially during startup or when feeding into a resistant process.
- **Reverse Rotation Capability:** Allows for convenient clearing of material jams, facilitating cleaning procedures, or reversing material flow if needed.
- **Accurate Dosing/Dispensing:** Enables precise volumetric or gravimetric dosing of materials, which is critical in applications like packaging (e.g., weigh-fill machines) or compounding processes.

Typical Drive Technologies Employed:

The choice of drive technology for an auger motor depends on the application's required precision, power, and cost considerations:

- **VFD (Variable Frequency Drive):** The most common choice for **AC motors**. VFDs offer excellent speed control, energy efficiency, and are robust enough for the continuous duty cycles often seen in auger applications.
- **Servo Drive:** Preferred for applications demanding **extreme precision and dynamic response**, particularly in high-accuracy dosing or weigh-fill machines where precise control of each "shot" or batch is critical.
- **Stepper Drive:** Often used for **low-power auger feeders** in smaller or less demanding machines where discrete, precise rotational steps are sufficient for dosing.

Optional Features & Advanced Control:

- **Pulse-based Speed Control:** Often integrated with **weight sensors (load cells)** for highly accurate gravimetric dosing, where the drive dispenses material based on weight rather than volume.
- **Timer Integration:** For intermittent or batch feeding applications, the drive can be integrated with timers to control the duration of auger rotation.
- **Manual/Auto Toggle:** Provides flexibility for operators to switch between manual control for setup/maintenance and automatic control during production.

5. Rewinder Drive –

A **Rewinder Drive** refers to the motor and drive system specifically designed to control the motor within a rewinder unit. These units are essential for rolling up processed materials—such as film, paper, foil, BOPP, nonwovens, and fabrics—onto a roll after they have undergone processes like printing, coating, lamination, or slitting.

Core Function:

The primary objective of the rewinder drive is to ensure that the material is wound onto the roll in a controlled and precise manner, guaranteeing a high-quality finished roll:

- **Smooth and Even Winding:** The drive must enable smooth and consistent winding to avoid uneven roll builds or material damage.

- **Proper Web Tension:** Maintaining the correct web tension throughout the winding process is crucial for preventing wrinkles, stretching, or misalignment of the material.
- **Synchronized Operation:** The rewinder drive must synchronize its speed and torque with the speed of the main processing line (e.g., laminator, extruder, printer) to ensure continuous and harmonious material flow.

Why a Dedicated Drive System is Essential for Rewinders:

The demands of precise and controlled roll-up necessitate a dedicated drive system for several key reasons:

- **Variable Speed Control:** As the roll diameter increases during winding, the rewinder's rotational speed must decrease to maintain a constant linear speed of the material. The drive must precisely manage this changing speed.
- **Constant Web Tension:** Maintaining consistent web tension is critical for preventing defects and ensuring a tightly wound, stable roll. The drive must dynamically adjust torque output to compensate for the changing roll diameter.
- **Synchronization with Main Line:** Seamless synchronization with the upstream processing line is vital to avoid material breakage (if rewinding too slowly) or slack (if rewinding too quickly).
- **Prevention of Material Damage:** Controlled acceleration and deceleration are essential to avoid sudden tension changes that could damage delicate materials.



Typical Drive Technologies Employed:

The selection of drive technology for a rewinder depends on the application's speed, precision requirements, and cost considerations:

- **AC Vector Drive / VFD (Variable Frequency Drive):** These are commonly used drives that offer excellent speed and torque control for rewinding applications. **Vector control** enables precise torque management, which is crucial for maintaining constant web tension.
- **Servo Drive:** Utilized in high-speed rewinding applications where **extreme precision, rapid dynamic response, and accurate positional control** are paramount. Servo systems offer superior performance for demanding applications.
- **Torque Motor Drive:** In simpler rewinder setups, a dedicated **torque motor** may be used. These motors are designed to provide high torque at low speeds and are often directly coupled to the rewinder shaft.

6. Edge Regrinding Drive –

An **Edge Regrinding Drive** refers to the motor and drive system specifically engineered to power the grinding or cutting mechanism responsible for trimming and processing **edge waste (side trims)** from continuous web materials. This critical process is common in industries handling films, foils, and sheets (such as BOPP, polyester, various plastics, and paper) after primary manufacturing or converting operations.

Core Function:

The primary function of the Edge Regrinding Drive is to facilitate the precise removal and processing of excess material from the edges of a web. This involves:

- **Trimming:** Precisely cutting off the uneven or unwanted side edges of the film or sheet.
- **Grinding/Granulating:** Reducing these trimmed edges into smaller, manageable particles.

- **Conveying:** Often, the processed material is then pneumatically conveyed or fed into a system for **recycling or direct reprocessing** (e.g., feeding back into an extruder).

Purpose of the Drive in Edge Regrinding:

A dedicated drive system for edge regrinding is essential for several reasons:

- **Rotation of Cutting/Grinding Mechanisms:** It provides the necessary power and controlled rotation for blades, grinding rollers, or other cutting elements to effectively trim and reduce the edge material.
- **Precise Edge Removal:** Ensures that the excess material is cut cleanly and consistently from the film or web edges, contributing to the quality and dimensional accuracy of the final product.
- **Synchronized Trimming Speed:** The drive often needs to operate at a speed precisely synchronized with the main production line's speed. This ensures continuous, accurate trimming regardless of changes in the web's linear velocity.
- **Handling Varying Loads:** The drive must be robust enough to handle fluctuating loads that arise from variations in material thickness, web speed, or the type of material being processed, ensuring consistent cutting performance.
- **Optimized Material Recovery:** By efficiently processing edge waste, the drive system supports efforts to minimize scrap and maximize material recovery for cost savings and environmental benefits.

Typical Drive Technologies Employed:

The choice of drive technology depends on the required precision, power, and specific grinding/cutting mechanism:

- **AC Induction Motor with VFD (Variable Frequency Drive):** This is the most common and robust solution. The VFD provides excellent speed control for



the motor, allowing the grinding/cutting speed to be precisely matched with the main line's speed and to handle varying material loads.

- **Servo Drive:** In highly precise or high-speed systems, a **Servo Drive** may be used. These offer superior dynamic response and positional accuracy, which can be beneficial for very fine or rapid trimming operations.
- **Direct Integration:** The drive system for edge regrinding is frequently **integrated with a blower or a granulator** that immediately sucks away and processes the trimmed material. This creates a compact and efficient waste handling subsystem.

7. Extruder Cart Drive –

An **Extruder Cart Drive** refers to the specialized motor and drive system responsible for controlling the **horizontal movement (forward and backward)** of an extruder machine or its barrel assembly. This movement typically occurs along a robust mechanical rail or cart system, facilitating critical operational tasks within various production lines.

Core Function:

The primary function of the Extruder Cart Drive is to enable the precise and controlled translation of the heavy extruder machinery. This mobility is essential for:

- **Alignment:** Accurately positioning the extruder head relative to the downstream die (e.g., in film blowing, sheet extrusion) or a lamination/coating station.
- **Maintenance:** Providing easy access to components for routine inspections, part replacements, or repairs.
- **Cleaning:** Facilitating the separation of the extruder from the die or other elements for thorough cleaning, especially during material changeovers or when clearing blockages.

Why a Dedicated Drive System is Essential for an Extruder Cart:

Given the substantial weight and critical positioning requirements of extruder machinery, a dedicated drive system offers significant advantages:

- **Precise Positioning:** Enables micro-adjustments to align the extruder head perfectly with the die or other processing equipment. This precision is vital for product quality and efficient operation.
- **Ease of Alignment:** Dramatically simplifies the complex and time-consuming task of aligning heavy machinery during startup, product changeovers, or when switching dies.
- **Reduced Manual Effort and Enhanced Ergonomics:** Eliminates the need for significant manual force or rudimentary lifting equipment, greatly reducing physical strain on operators and improving workplace safety.
- **Safe and Smooth Movement:** Ensures that large and heavy equipment moves in a controlled, stable, and jolt-free manner, minimizing the risk of mechanical damage to the machinery itself or injury to personnel.
- **Increased Efficiency:** Reduces downtime associated with manual adjustments, maintenance, and cleaning cycles, thereby improving overall production efficiency.

Typical Components and Application:

An Extruder Cart Drive system typically consists of:

- **Motor:** Often an AC induction motor, chosen for its robustness and power.
- **Drive:** An AC Variable Frequency Drive (VFD) is commonly used to provide precise speed and torque control to the motor.
- **Gearbox:** To provide the necessary torque multiplication for moving heavy loads.
- **Mechanical System:** Rails, wheels, and a cart assembly specifically designed to bear the extruder's weight and facilitate smooth horizontal motion.



Core Parameters for Drive Configuration-

S NO .	PARAMETER NO.	NAME	EXTRUDER PARAMETER	LAMINATOR PARAMETER	REWINDER PARAMETER	EDGE REGRINDING PARAMETER
1	1	Maximum Frequency	50	50	50	50
2	2	Minimum Frequency	0	0	0	0
3	3	Base Frequency	50	50	50	50
4	4	Multi Speed Setting(High Speed)	50	50	50	50
5	5	Multi Speed Setting(Middle Speed)	30	30	30	30
6	6	Multi Speed Setting(Low Speed)	10	10	10	10
7	7	Acceleration Time	15	60	20	20
8	8	Deceleration Time	15	10	8	8
9	9	Electronic Thermal O/L relay/ Rated Motor Current	Rated Motor Current	Rated Motor Current	Rated Motor Current	Rated Motor Current
10	10	DC injection Brake Operation Frequency	3	3	3	3
11	11	DC injection Brake Operation Time	0.5	0.5	0.5	0.5
12	13	Starting Frequency	0.5	0.5	0	0
13	18	High Speed Maximum Frequency	50	50	55	55
14	19	Base Frequency Voltage	8888	8888	8888	8888
15	20	Acceleration/ Deceleration reference Frequency	50	50	50	50
16	71	Applied Motor	3	3	3	3
17	73	Analog Input Selection	0	0	0	0
18	75	Reset Selection/ Disconnected PU Detection/ PU Stop Selection/ Reset Limit	1	1	1	1
19	77	Parameter Write Selection	2	2	2	2
20	79	Operation Mode Selection	2	2	2	2
21	80	Motor Capacity	Rated Motor KW	Rated Motor KW	Rated Motor KW	Rated Motor KW
22	81	Number Of Motor Poles	Rated Motor Poles	Rated Motor Poles	Rated Motor Poles	Rated Motor Poles
23	83	Rated Motor Voltage	Rated Motor Voltage	Rated Motor Voltage	Rated Motor Voltage	Rated Motor Voltage
24	84	Rated Motor Frequency	Rated Motor Frequency	Rated Motor Frequency	Rated Motor Frequency	Rated Motor Frequency

25	195	ABC1 Terminal Function Selection	11	11	11	11
26	250	Stop Selection	9999	9999	9999	9999
27	267	Terminal Four Input Selection	0	0	2	2
28	331	RS-485 Communication Station Number	2	3	4	5
29	332	RS-485 Communication Speed	96	96	96	96
30	333	RS-485 Communication Stop Bit Length/ Data Length	0	0	0	0
31	334	RS-485 Communication Parity Check Selection	2	2	2	2
32	335	RS-485 Communication Retry Count	3	3	3	3
33	336	RS-485 Communication Check Time Interval	5	5	5	5
34	337	RS-485 Communication Waiting Time Setting	9999	9999	9999	9999
35	338	Communication Operation Command Source	1	1	1	1
36	339	Communication Speed Command Source	1	1	1	1
37	340	Communication Startup Mode Selection	0	0	0	0
38	341	RS-485 Communication CR/LF Selection	1	1	1	1
39	549	Protocol Slection	1	1	1	1
40	551	PU Mode Operation Command Source Selection	2	2	2	2

Parameter NO.	Name	AUGER PARAMETER S	EXTRUDER CART PARAMETER	MELT PUMP PARAMETER
1	Maximum Frequency	50	50	50
2	Minimum Frequency	0.2	0.2	0.2
3	Base Frequency	50	50	50
4	Multi Speed Setting(High Speed)	50	50	50
5	Multi Speed Setting(Middle Speed)	30	30	30
6	Multi Speed Setting(Low Speed)	10	10	10
7	Acceleration Time	10	10	10
8	Deceleration Time	10	10	10
9	Electronic Thermal O/L relay/ Rated Motor Current	Rated Motor Current	Rated Motor Current	Rated Motor Current
10	DC injection Brake Operation Frequency	0.01	0.01	0.01

11	DC injection Brake Operation Time	0.1	0.1	0.1
13	Starting Frequency	0.01	0.01	0.01
18	High Speed Maximum Frequency	50	50	50
19	Base Frequency Voltage	415	415	415
20	Acceleration/ Deceleration reference Frequency	50	50	50
71	Applied Motor	3	3	3
73	Analog Input Selection	0	0	0
75	Reset Selection/ Disconnected PU Detection/ PU Stop Selection/ Reset Limit	1	1	1
77	Parameter Write Selection	2	2	2
79	Oeration Mode Selection	2	2	2
80	Motor Capacity	Rated Motor KW	Rated Motor KW	Rated Motor KW
81	Number Of Motor Poles	Rated Motor Poles	Rated Motor Poles	Rated Motor Poles
83	Rated Motor Voltage	Rated Motor Voltage	Rated Motor Voltage	Rated Motor Voltage
84	Rated Motor Frequency	Rated Motor Frequency	Rated Motor Frequency	Rated Motor Frequency
117	PU Communication Station Number	6	7	8
118	PU Communication Speed	96	96	96
119	PU Communication Stop Bit Length	0	0	0
120	PU Communication Parity Check	2	2	2
121	Number Of PU Communication Retries	3	3	3
122	PU Communication check Time Interval	4	4	4
123	PU Communication Waiting Time Setting	0	0	0
124	PU Communication CR/LF Selection	1	1	1
125	Terminal 2 Frequency Setting Gain Frequency	50	50	50
160	Extended Function Display Selection	0	0	0
181	RM Terminal Function Selection	62	62	62
192	ABC Terminal Function Selection	11	11	11
250	Stop Slection	9999	9999	9999
338	Communication Operation Command Source	0	0	0
339	Communication speed Command Source	0	0	0
340	Communication Start up Mode Selection	0	0	0
502	Stop Mode Selection at Communication Error	1	1	1
549	Protocol Selection	1	1	1
551	PU Mode Operation Command Source Selection	9999	9999	9999

Mastering Cable Detail Reports –

During my internship, one of the most crucial skills I honed was preparing comprehensive **cable detail reports** for industrial automation systems. This meticulous process involves thoroughly documenting each cable's journey, type, core specifications, usage, and other vital parameters. These detailed reports are indispensable, serving as the backbone for both efficient installation and seamless future maintenance of complex industrial wiring.

The Paramount Importance of Accurate Cable Documentation

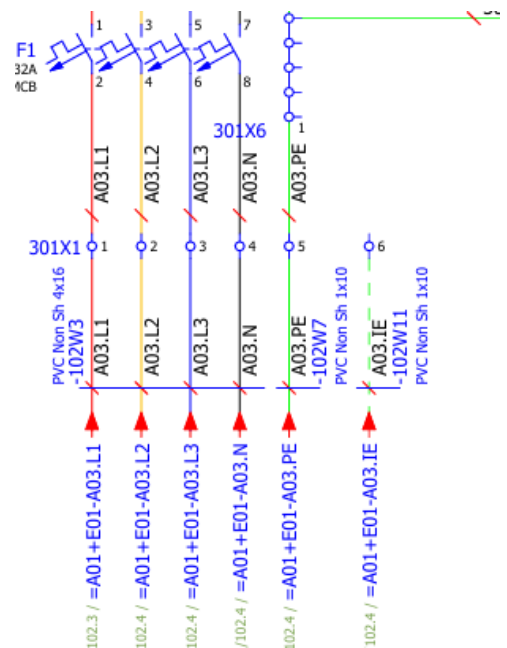
Cable detailing is the bedrock of robust industrial wiring. It encompasses the precise documentation of every cable connecting critical devices such as sensors, drives, PLCs, junction boxes, motors, and control panels. Without accurate and exhaustive cable details, the entire wiring infrastructure can devolve into an indecipherable maze, rendering troubleshooting and modifications incredibly challenging.

Tracing Connections: The "From – To" Methodology

A fundamental aspect of cable detailing I mastered was accurately specifying the **"From – To"** path for each cable. This defines the exact origin and destination, providing an unambiguous connection roadmap. For instance:

- **From:** PLC Panel – Terminal TB1
- **To:** Sensor at Machine A – JB1

This standardized format offers immediate clarity on the cable's connection trajectory, proving invaluable for electricians and engineers during installation. I also learned the importance of including **intermediate boxes**, such as Junction Boxes (JBs), if a cable traverses through them, ensuring a complete and accurate route description.



Demystifying Cable Types and Specifications

My training introduced me to a diverse array of cable types and the proper methods for documenting their specifications. A common notation I frequently encountered was:

0.5 mm² × 16 This notation signifies:

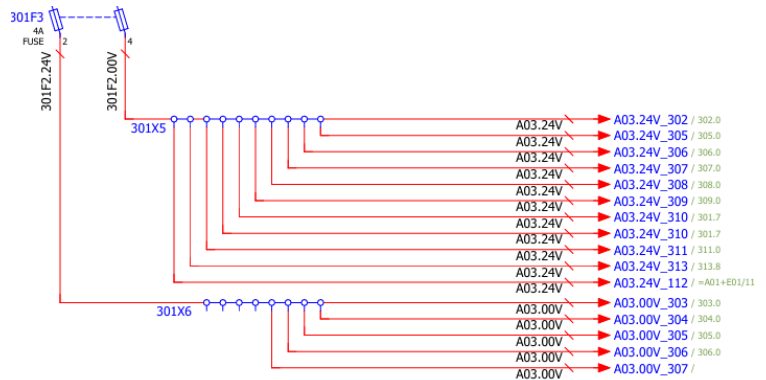
- **0.5 mm²**: The **cross-sectional area** of each individual core (wire).
- **× 16**: The cable contains **16 individual cores**.

Such cables are typically deployed for low-power control signals within automation systems. Beyond these, I also gained familiarity with:

- **Power cables** (e.g., 4×6 mm²)
- **Communication cables** (e.g., shielded twisted pair)
- **Instrumentation cables**

Precision in Core Identification: Numbering and Color Coding

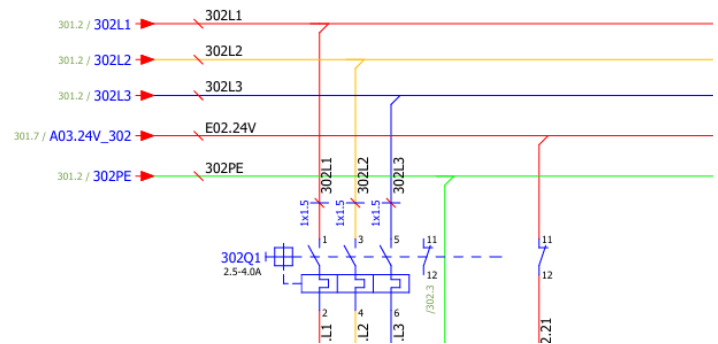
Accurate identification of each wire (core) within a cable is paramount, achieved through either **color coding** or **numbering**. I learned to meticulously document core numbers, their specific functions, and their ultimate termination points. For example:



Core No.	Function	Termination
1	+24V Supply	PLC TB1 - Terminal 1
2	Digital Output	JB1 - Terminal 2

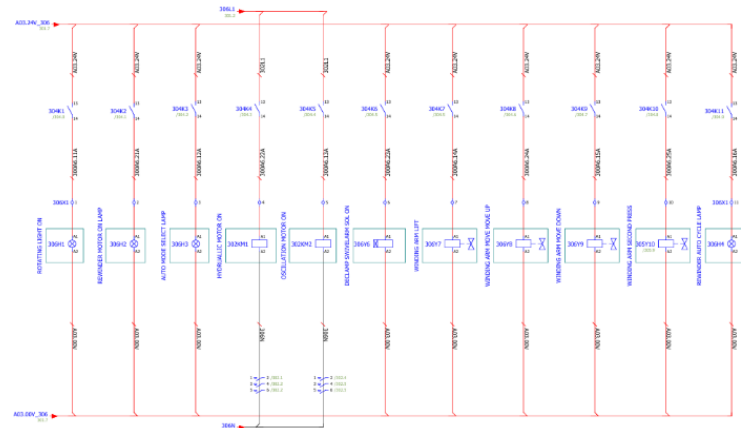
Ensuring Traceability: Ferrule Printing and Tagging

A key skill I developed was utilizing **ferrule printers** for tagging individual wires. Each wire receives a unique, descriptive label (e.g., DO1, AI2, +24V) to ensure easy identification during wiring and future troubleshooting. The utmost precision is required here; these ferrule tags must **exactly match** the specifications in the cable schedule and wiring diagrams.



Beyond the Technical: Cable Routing and Dressing

My learning extended beyond technical specifications to the critical aspects of **physical cable management**. This included understanding proper cable routing techniques, the strategic use of cable trays or ducts, and the importance of neatly tying cables. These practices not only enhance the aesthetic appeal but also significantly improve safety and facilitate future inspections.



Conclusion: A Foundation for Future Success

In essence, preparing cable detail reports provided me with an invaluable understanding of how an entire industrial wiring system is meticulously planned and documented. This experience cultivated a sharp **attention to detail**, enhanced my proficiency in **interpreting complex wiring diagrams**, and fostered crucial **real-time coordination skills** with both control panels and field devices. This foundational knowledge will undoubtedly be a significant asset in my future endeavors within industrial automation, panel design, and maintenance roles.

Programming & Automation The Foundation of Modern Industrial Systems

Programming and automation are two inextricably linked concepts that form the bedrock of contemporary industrial operations. Together, they drive efficiency, accuracy, and safety in manufacturing and various other sectors.

Programming in this context refers to the meticulous process of writing logic-based code that dictates the precise behavior of industrial machinery. This code is typically developed for specialized controllers such as **Programmable Logic Controllers (PLCs)** or **microcontrollers**. Common programming languages used in industrial settings include:

- **Ladder Diagram (LD):** A graphical language that visually resembles the rungs of a ladder, popular for its intuitive representation of control logic.
- **Structured Text (ST):** A high-level, text-based language similar to Pascal, offering more complex algorithmic capabilities.
- **Function Block Diagram (FBD):** A graphical language that uses pre-defined function blocks to represent control functions, simplifying complex operations.

The purpose of this programming is to define every aspect of machine operation, from simple on/off commands to intricate sequential processes and real-time control.

Automation, on the other hand, encompasses the broader application of control systems to operate machinery with minimal human intervention. It leverages a suite of technologies and components to achieve autonomous or semi-autonomous operation, including:

- **PLCs (Programmable Logic Controllers):** The "brains" of many automated systems, executing programmed logic to control outputs based on inputs.
- **Sensors:** Devices that detect physical parameters (e.g., proximity, temperature, pressure) and convert them into electrical signals for the controller.
- **Drives:** Electronic devices that control the speed, torque, and position of motors, enabling precise motion control.

- **HMI (Human-Machine Interfaces):** Graphical user interfaces that allow operators to monitor and interact with the automated system, providing data visualization and control capabilities.

The synergy between programming and automation is what truly transforms industrial processes. Programming provides the intelligence and instructions, while automation implements these instructions through a coordinated system of hardware and software. This combined approach yields significant benefits, including:

- **Increased Productivity:** Machines can operate continuously and at higher speeds than manual processes.
- **Enhanced Accuracy and Consistency:** Automated systems perform tasks with repeatable precision, reducing errors and ensuring uniform product quality.
- **Improved Safety:** By taking over hazardous or repetitive tasks, automation minimizes human exposure to dangerous environments and reduces the risk of accidents.

In essence, programming provides the "how-to" guide, and automation provides the "doing" mechanism, ensuring that industrial machines operate consistently and efficiently to meet modern production demands.

Programming a Turret Motor for Industrial Automation –

During my internship, I gained invaluable hands-on experience in **motor control programming**, specifically focusing on a **Turret Motor** commonly used in industrial automation systems. My primary objective was to program the turret motor to achieve precise positional movements based on complex control logic. While the company primarily utilizes **Automation Studio** for its development, I had the unique opportunity to work with **Flexi Soft**, which provided a deeper understanding of logic-based motor control and significantly enhanced my skills in industrial automation and programming.

Understanding the Turret Motor's Functionality

A turret motor's primary role is to accurately **rotate a machine part to specific angular positions**, often found in applications like packaging or labeling. Depending on the operation, this requires both **position-based control** and, at times, **speed regulation**. My tasks involved:

- Programming **forward and reverse motion**.
- Implementing precise **stop conditions**.
- Integrating with **sensors** for homing (initial reference position) or end-limit detection.
- Developing logic for robust **safety interlocks**.
- Successfully completing these tasks necessitated a solid grasp of various motor types, the functionality of limit switches, and adherence to critical safety protocols.

Programming with Flexi Soft

Flexi Soft, a safety controller platform developed by SICK, proved to be an excellent tool for integrating both control logic and safety functionalities. My work with Flexi Soft involved:

- Creating intricate **logic blocks** using its intuitive graphical programming interface.
- Defining and configuring various **input/output (I/O) connections**.
- Monitoring real-time **motor feedback** via integrated sensors.
- Implementing critical **emergency stop logic** and establishing safety zones.
- A key learning outcome was mastering the application of **Boolean logic (AND, OR, NOT)** to construct sophisticated motor run conditions, ensuring the turret only operated when all safety parameters were unequivocally met. Although Automation Studio is the company's primary machine automation platform, my experience with Flexi Soft provided a robust foundation in logic-based safety and control, skills that are highly transferable across different industrial environments.

Flexi Soft vs. Automation Studio: A Comparative Insight

Automation Studio (from B&R or Famic Technologies, depending on the specific implementation) is a more advanced, PLC-integrated development environment designed for creating comprehensive automation sequences, encompassing motion control, Human-Machine Interface (HMI) development, and communication protocols. My experience allowed for a direct comparison of the two platforms:

Feature	Flexi Soft	Automation Studio
Focus Area	Safety and basic logic control	Complete machine automation
Programming	Block/logic diagrams	Ladder, Function Block Diagram (FBD), Structured Text
I/O Control	Safety I/O, Interlocks	Motion control, Servo tuning, general I/O
Company Usage	Used selectively for safety-specific tasks	Main platform for machine development

While Flexi Soft is specialized for safety and fundamental logic, the underlying logical structures I developed can be directly translated and integrated into Automation Studio using FBD or Ladder Logic, providing enhanced control and system integration.

The Programmer's Edge: Unlocking Diverse Skills Through Code

This programming task significantly enhanced several key skills essential for a career in industrial automation:

- Sharpened **logical thinking** and practical application of Boolean control logic.
- A deeper understanding of **motor interlocks** and complex **feedback systems**.
- Hands-on experience with industrial **safety standards** and their implementation.
- Proficiency in **interpreting electrical wiring and I/O diagrams**.
- The ability to **bridge conceptual understanding** between diverse programming platforms (Flexi Soft vs. Automation Studio).
- This experience instilled confidence in my ability to independently program motion-based systems and provided critical insight into the indispensable role **safety logic** plays in real-time industrial automation. It also helped me connect the foundational principles learned on one platform (Flexi Soft) to broader, more integrated systems (Automation Studio), knowledge that will be invaluable for future work in PLC programming, industrial safety, and machine design.

HMI Development in B&R Automation Studio

During my internship, a pivotal area of my work was **Human-Machine Interface (HMI) development** using B&R's **Automation Studio**. The HMI serves as the critical bridge between the machine and the human operator, facilitating interaction through intuitive displays, interactive buttons, and clear alarm indications. My focus was on developing comprehensive and interactive screens for a **Turret Motor Control System**, encompassing navigation, input fields, real-time feedback, and alarm management. This experience significantly enhanced both my software development and industrial automation skills.

While my initial exposure during the internship involved **Flexi Soft**—a platform primarily focused on safety and logic—Automation Studio provided the unique opportunity to build a fully visual interface seamlessly integrated with PLC logic, offering a more complete and holistic automation development environment.

Understanding HMI: Its Importance and Role in Automation Studio

HMI Overview

An HMI refers to the graphical screens that empower operators to interact with a machine, visualize system data, and issue commands. In contemporary industrial systems, HMIs are indispensable for:

- **Monitoring real-time data** and operational parameters.
- **Issuing commands** such as start, stop, and reset.
- Providing clear **alarm indications** and aiding in troubleshooting.
- Allowing operators to **change setpoints** and configuration parameters.



Role in Automation Studio –

Within Automation Studio, HMI development is managed under the **Visualization Manager**. This module directly links the graphical interface elements to the PLC code and its defined tags. I leveraged this feature to:

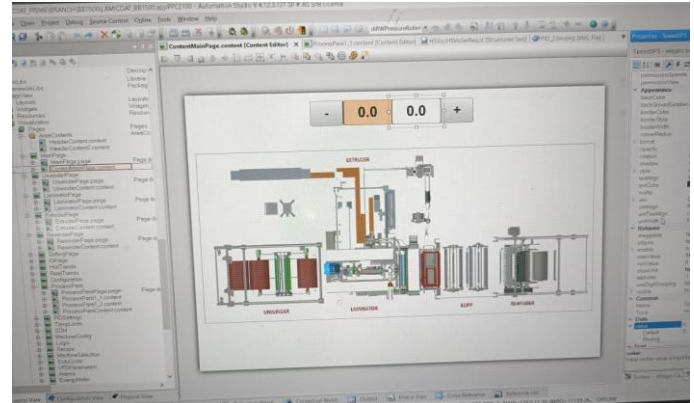
- Design and create **functional screens**.
- Incorporate **real-time feedback indicators** for dynamic data visualization.
- Program **user-defined navigation** paths between screens.
- Integrate comprehensive **alarm and diagnostic displays**.

Steps to Create an HMI in Automation Studio

Planning and Mockup Creation

Before commencing development, I initiated the process by creating a **mockup layout**. This planning phase was crucial for ensuring a logical user interface (UI) flow and organized screen structure. My mockup typically included:

- A central **Home Screen**.
- A dedicated **Turret Control Panel**.
- An **Alarm Page** for notifications.
- A **Settings and Diagnostics Page** for configuration.



Designing Buttons and Implementing Navigation Logic –

Adding Buttons

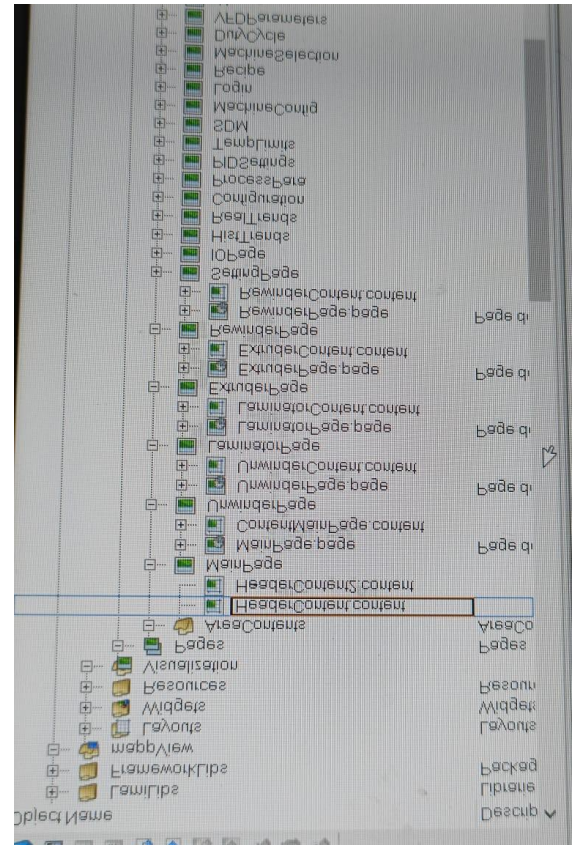
The process of adding buttons in Automation Studio is straightforward:

1. Open the desired visualization page.
2. Drag a "Button" object from the toolbox onto the screen.
3. Configure its properties, including:
 - **Label:** e.g., "Start Turret"
 - **Assigned Action:** Link to a Boolean variable, Integer, String, or Character data type.
 - **OnClick Trigger:** Define the specific action to be executed when the button is pressed.

Types of Buttons Utilized

I worked with various button types, each serving a distinct purpose:

- **Standard Button:** Triggers a single, momentary action (e.g., Start, Stop, Reset).
- **Toggle Button:** Switches between ON/OFF states with each press (e.g., Motor ON/OFF).
- **Navigation Button:** Facilitates screen transitions using the PageCall() function.
- **Multi-Function Button:** Executes multiple actions simultaneously (e.g., setting a variable, triggering an alarm, and changing the page).



Assigning Data Types to Buttons for Dynamic Control

Buttons are intrinsically linked to PLC variables, which are defined with specific data types to manage machine behavior.

Boolean / Integer Examples

For actions like starting a motor, a **Boolean variable** (bStartMotor) is typically used. The HMI button's OnClick action is bound to this variable. Within the PLC code, this might translate to:

```
// Tag: bStartMotor : BOOL
```

```
IF bStartMotor THEN
```

```
Motor_State := TRUE;
```

```
END_IF;
```

Similarly, **Integer variables** can represent multiple states. For instance, a button might write 1 to an integer tag (iTurretMode) to set the turret to "Auto" mode, simplifying complex logic handling in the PLC:

```
// Tag: iTurretMode : INT
```

```
// 0 = Manual, 1 = Auto, 2 = Maintenance
```

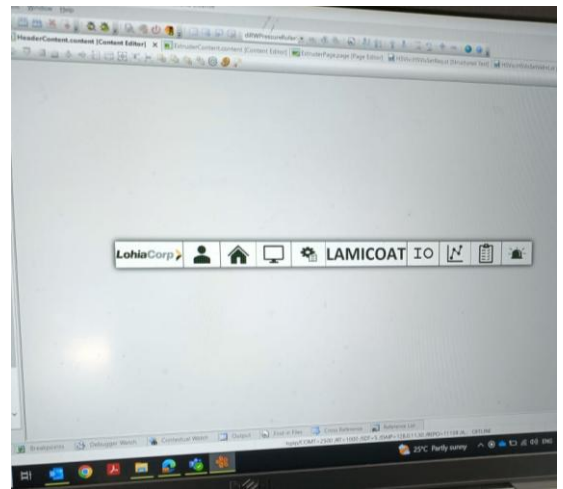
String / Character Examples –

For buttons that modify operational modes or display messages, **String or Character variables** are used.

For example, a button could be configured to write 'AUTO' to a string tag (sMode) when pressed:

```
// Tag: sMode : STRING[10]
```

```
sMode := 'AUTO';
```



Seamless Screen Navigation –

Effective navigation is crucial for user experience. I implemented various methods to switch between HMI pages:

Basic Configuration

To switch between pages, the PageCall("ScreenName") function is configured under a button's properties.

Conditional Navigation

For more dynamic control, conditional navigation ensures users access relevant screens based on specific criteria (e.g., user access level):

```
IF iUserLevel = 1 THEN
```

```
PageCall("AdvancedSettings");
```

```
ELSE
```

```
PageCall("BasicView");
```

```
END_IF;
```

Intuitive Image-Based Navigation –

A highly effective technique involved using a background image (e.g., a machine layout) and overlaying transparent buttons or **hotspot areas**. Each hotspot was linked to a different navigation target, such as:

- **Motor area** → Motor Control Page
- **Sensors area** → Sensor Status Page
- **Alarm icon** → Alarm Page



This approach created a visually intuitive UI akin to a smart factory layout map.

Robust Alarm Configuration –

Alarm management is a cornerstone of safe and efficient HMI operation. I configured various types of alarms in Automation Studio:

Types of Alarms

- **Digital Alarms:** Triggered by Boolean variables (e.g., Motor Overload).
- **Analog Alarms:** Activated when a value crosses a defined threshold (e.g., Temperature > 60°C).
- **Custom Messages:** Display operator-friendly text based on specific PLC logic.

Alarm Handling and Display

Alarms are defined within the **Project > System > Alarm Configuration** section, where parameters such as the triggering tag (e.g., bOverload), a descriptive message, the alarm

class (Warning, Error, Critical), and acknowledgement actions are set. I also implemented a dynamic pop-up page to display active alarms:

```
IF bAlarm THEN
```

```
PageCall("AlarmPopup");
```

```
END_IF;
```

Advanced Monitoring: Trend Viewer and Data Logging –

For in-depth analysis and historical tracking, I utilized the **Trend Viewer** to plot real-time data such as motor speed and temperature. This involved linking the X-axis to time and the Y-axis to relevant tags (e.g., iMotorSpeed). Additionally, I enabled **data logging** for historical review and export capabilities.

Sample Main PLC Code (for HMI Linkage) –

The HMI's functionality is directly tied to the underlying PLC logic. A simplified example illustrating this linkage is:

```
VAR
```

```
bStartMotor    : BOOL;
```

```
bStopMotor     : BOOL;
```

```
iTurretMode    : INT;
```

```
sCurrentStatus : STRING[20];
```

```
bAlarm         : BOOL;
```

```
END_VAR
```

```
IF bStartMotor THEN
```

```
Motor_State    := TRUE;
```

```
sCurrentStatus := 'Running';
```

```
END_IF;
```

```
IF bStopMotor THEN
```

```
Motor_State := FALSE;
```

```
sCurrentStatus := 'Stopped';
```

```
END_IF;
```

```
IF Motor_Overload THEN
```

```
bAlarm := TRUE;
```

```
END_IF;
```

This code directly interacts with the buttons, indicators, and alarm displays on the HMI screen.

Key Learning Outcomes –

Technical Skills Gained

- Proficiency in creating professional and intuitive **HMI interfaces**.
- Expertise in configuring diverse **button types** and linking them to various **data types**.
- Competence in handling **alarms, navigation, and password protection**.
- Practical experience with **tag databases** and **ENUM mappings**.

Practical Understanding

Developed a keen eye for **UI design** tailored to enhance the operator experience.

Mastered **navigation linking** based on practical machine layouts.

Gained a deeper understanding of **safety interlocks** and **feedback handling** in an integrated system.

Challenges and Solutions

Like any complex development task, HMI creation presented its challenges:

Managing numerous screens: I addressed this by organizing screens into **page folders** for better management.

Tag synchronization delays: This was mitigated by consistently **rebuilding the project** before testing tag functionalities.

Understanding conditional logic for navigation: I overcame this by creating detailed **screen flow diagrams** to visualize and plan complex navigation paths.

Conclusion –

My HMI development experience in Automation Studio provided a comprehensive, **full-stack view of industrial automation**—from the underlying control logic to the end-user interaction. I learned to program interactive, functional interfaces that communicate directly with PLCs, enabling real-time control and monitoring. The ability to assign variables, navigate across multiple pages, utilize trends and alarms, and construct a user-friendly layout has significantly boosted my confidence in handling real-world automation systems.

This experience also clarified the comparative limitations of platforms like Flexi Soft, which, while excellent for logic and safety, do not offer full-scale HMI capabilities. Automation Studio, conversely, provides a complete **control + interface solution**, cementing its position as a powerful and indispensable tool for any aspiring industrial automation engineer.

Challenges Faced And their Solution

Working across our diverse fleet of lamination and printing machines – including the automated BB1500 and semi-manual C1500 laminators, and the Prismaflex 140 and Prismaflex CI printing machines – has presented a wide spectrum of operational and technical challenges. Each issue, from intricate electrical rewiring to HMI diagnostics and precise mechanical path corrections, demanded a deep understanding of industrial systems, advanced automation components, and complex process flows.

Below are the major issues encountered and the solutions implemented across these varied machines:

Lamination Machines: BB1500 (Automated) & C1500 (Semi-Manual)

The lamination section involves managing two distinct operational philosophies: the high-precision automation of the BB1500s and the operator-dependent nature of the C1500s.

Common Challenges & Solutions (across both BB1500 and C1500):

- **Issue: Web Misalignment and Wrinkling:**

Root Cause: Inconsistent tension, worn-out rollers, or improper web guiding. More prevalent on C1500 due to manual control.

Solution: For BB1500, recalibrated tension sensors and adjusted PID loops in Mitsubishi VFDs. For C1500, implemented regular inspection and replacement of idler rollers, and provided enhanced operator training on manual tension adjustments and web path correction.

- **Issue: Poor Adhesion/Delamination:**

Root Cause: Incorrect temperature, pressure, or inadequate dwell time. Could also be due to material incompatibility.

Solution: Calibrated heater temperature sensors on both machines. For BB1500, optimized pressure control parameters through HMI. For C1500, established clear operational

guidelines for pressure roller settings and conducted material compatibility tests to ensure optimal bonding.

- **Issue: Motor Overload/Tripping:**

Root Cause: Mechanical friction, electrical imbalances, or incorrect VFD parameter settings. More critical on BB1500 due to integrated drive systems.

Solution: Diagnosed and corrected electrical rewiring issues. Adjusted current limits and acceleration/deceleration ramps in Mitsubishi VFDs for BB1500. Lubricated mechanical components and checked motor bearings on both machines.

Specific Challenges & Solutions (BB1500 Automated):

- **Issue: HMI Communication Errors / Control Glitches:**

Root Cause: Software bugs, faulty communication cables, or VFD firmware issues.

Solution: Performed HMI diagnostics and updated firmware where necessary. Replaced damaged Ethernet/communication cables and reconfigured network settings for seamless data exchange between sensors, VFDs, and the HMI.

- **Issue: Inaccurate Tension Regulation (Automated System):**

Root Cause: Malfunctioning load cells or drift in VFD tension control algorithms.

Solution: Re-calibrated load cells and fine-tuned tension control algorithms within the Mitsubishi VFDs, optimizing responsiveness and stability.

Specific Challenges & Solutions (C1500 Semi-Manual):

- **Issue: High Scrap Rate due to Operator Error:**

Root Cause: Lack of consistent manual control, fatigue, or insufficient training.

Solution: Developed detailed standard operating procedures (SOPs) for manual unwinding, rewinding, and tension control. Implemented a robust training program emphasizing hands-on practice and quality checks at each stage.

- **Issue: Inconsistent Speed and Output:**

Root Cause: Manual speed adjustments not precise enough for demanding jobs.

Solution: While fundamentally manual, identified specific manual control points for enhancement, such as marking optimal speed settings for common materials, to guide operators for better consistency.

Printing Machines: Prismaflex 140

The Prismaflex printing machines introduce complexities related to color registration, ink management, and precise material handling for high-quality print output. The "CI" in Prismaflex CI likely refers to Central Impression, implying a drum-based system for improved registration.

Common Challenges & Solutions (across both Prismaflex 140 and Prismaflex CI):

- **Issue: Poor Print Registration:**

Root Cause: Mechanical wear in gears or bearings, inconsistent web tension, or issues with plate mounting.

Solution: Conducted detailed mechanical inspections to identify and rectify worn components. Optimized tension control systems (manual or automated, depending on the model) to ensure consistent web travel. Implemented stricter procedures for mounting and aligning printing plates.

- **Issue: Ink Quality and Consistency Issues (e.g., streaking, uneven color):**

Root Cause: Improper ink viscosity, contaminated ink, or issues with anilox rollers/doctor blades.

Solution: Established a rigorous ink mixing and viscosity control protocol. Regularly cleaned and maintained anilox rollers and doctor blades. Investigated and addressed any potential contamination sources in the ink supply system.

- **Issue: Web Breaks/Tears during Printing:**

Root Cause: Excessive tension, sharp edges on machine components, or inherent material flaws.

Solution: Reviewed and optimized tension settings across all printing zones. Identified and deburred any sharp edges or rough surfaces along the web path. Implemented stricter incoming material quality checks.

- **Issue: Speed Limitations for High-Volume Jobs:**

Root Cause: Design limitations of the machine, potentially without the benefits of a central impression drum for stability at higher speeds.

Solution: Optimized setup times and minimized changeover durations to maximize available running time. Focused the Prismaflex 140 on jobs that are less speed-sensitive or require shorter runs, while directing high-volume work to the Prismaflex CI where feasible.

Troubleshooting the BB1500 Lamination Machine –

Working with the BB1500 lamination machine, a high-performance industrial system, involves intricate troubleshooting across its electrical, automation, and mechanical components. The challenges faced often require a blend of theoretical knowledge and practical, hands-on application.

1. Rewinder Diameter Exceeded Limit & Declamp Failure

Challenge: During operation, the rewinder roller surpassed its predefined maximum diameter, preventing the automatic declamp function from engaging. This was an HMI-controlled logic issue, driven by feedback from the diameter counter or encoder.

Solution:

- **Diagnosed the limit logic** configured within the HMI.
- **Accessed rewinder diameter setting parameters** and verified the actual value against the setpoint threshold.
- After confirmation, the value was **recalibrated**, and the declamp interlock condition was **temporarily edited via HMI and PLC** to allow for a manual override.
- Once rewinding was safely completed, the **original limits were restored**, and the declamp sensor was thoroughly tested to verify its triggering range.

2. Cable Sensor Load Absence (Internal 0V from PLC)

Challenge: Two cable sensors for presence detection failed to register any load, meaning no signal reached the PLC. Upon inspection, the sensors were not receiving a correct ground reference.

Solution:

- Instead of relying on an unreliable or floating ground, the **sensor signal was connected directly to the PLC's internal 0V**, which provided a stable common reference.

- This immediately resulted in **proper signal reception** at the input terminals.
- The sensors' triggering was thoroughly tested using manual presence simulation to confirm consistent operation.

3. HMI-Based Troubleshooting of Rewinder Pressure System

Challenge: The BB1500's rewinder section exhibited inconsistent pressure buildup, negatively impacting winding tension and final lamination quality. The issue appeared pneumatic but without visible leakage.

Solution:

- Accessed the **HMI pressure monitoring screen** to check real-time feedback from the analog pressure sensors.
- The **PID control loop was analyzed** for improper setpoint or gain values.
- After trial adjustments, the **pressure setpoint was finely tuned**, and deadband margins were slightly increased.
- The **control valve response time was logged and tested**.
- **Recalibration of the pressure sensor** was performed using a known air supply reference.

4. Edge Regrinding Circuit Troubleshooting

Challenge: The edge regrinding system unexpectedly ceased operation. Despite the motors and cutter mechanism being physically intact, the circuit failed to respond to any commands.

Solution:

- Opened the electrical cabinet and **traced the control circuit** of the edge regrinding module.
- Discovered that a **control relay was not receiving a proper 24V signal** due to a damaged wire.

- The faulty section was **replaced**, and the system was retested using a manual jog function to verify proper response.
- The **circuit diagram was updated** accordingly to reflect the corrected path.

5. Motor Direction Reversal Using Star-Delta at Main MCB

Challenge: One of the system's motors (likely for the rewinder or unwinder) was running in the opposite direction, causing incorrect material pull and tension issues.

Solution:

- Since the motor utilized a Star-Delta starter, the most efficient solution was to **interchange two motor phases at the Main Circuit Breaker (MCB) output terminal**.
- This effectively **reversed the motor direction** without altering any drive parameters.
- **Markings were updated** to indicate the new connection, and motor protection settings were rechecked.

6. Fabric Tension Incorrect Due to Missing Rollers in the Path

Challenge: Inconsistent tension across the lamination fabric was observed. Upon inspection, several guide rollers were found to be missing from the material path setup.

Solution:

- The correct roller path was **reconstructed using the machine layout manual** and old drawings.
- The **missing guide rollers were installed**, ensuring even material distribution.
- The fabric web was re-threaded, and the **tension control from the VFD and dancer roller was thoroughly tested**.
- Post-correction, a significant improvement in final laminated quality was observed.

7. Earthing for Current Transformer (CT) for Energy Meter Check

Challenge: During energy measurement and system calibration, the Current Transformer (CT) output was not grounding properly, leading to erratic energy meter readings.

Solution:

- In a Star connection system, an **artificial neutral was introduced between the phases using resistors** to create a proper earth reference for the CT.
- This **stabilized the CT output**, enabling consistent and correct readings from the energy meter.
- A **continuity check and earth loop impedance test** confirmed the safe installation.

8. Roller Direction Was Wrong – Reversed to Apply Fabric Tension

Challenge: The machine was feeding fabric with incorrect tension due to a roller rotating in the wrong direction, resulting in loose fabric before the lamination stage.

Solution:

- The **incorrect wiring or drive direction** for the roller motor was identified.
- The direction was **reversed either by a drive parameter change or by interchanging wiring phases**.
- The **roller orientation was also checked** to ensure correct material flow.
- After correction, fabric tension significantly improved, and proper alignment was achieved.

Each of these challenges on the BB1500 machine demanded a practical, multi-faceted approach, integrating electrical knowledge, automation control, mechanical alignment, and a deep understanding of the overall process flow. By systematically diagnosing issues across sensors, Human-Machine Interfaces (HMI), Programmable Logic Controller (PLC) inputs, motor control, and pneumatic systems, full operational capacity was successfully restored with minimal downtime.

Acknowledging a Profound Internship

Experience

My internship with the BB1500 lamination line has concluded, marking an exceptionally valuable phase in my professional development. This period provided me with a comprehensive, hands-on immersion into the intricacies of industrial automation and electrical systems, significantly deepening my understanding and practical capabilities.

Key Contributions and Technical Growth

During my tenure, I actively engaged with critical aspects of the BB1500 machine's lifecycle and operation. My responsibilities and learning opportunities included:

- **Electrical System Design & Documentation:** Spearheading the creation of electrical Bills of Materials (BOM) using **EPLAN** and thoroughly mapping the machine's complete circuit diagrams, ensuring clarity and precision for future reference.
- **Wiring and Connectivity:** Gaining in-depth knowledge of the floor wiring layout and its practical, real-world connections, crucial for robust system integration.
- **Pneumatic Systems Expertise:** Understanding and applying principles of pneumatic circuits and their operational sequences within the lamination process.
- **Process Optimization:** Analyzing the operational parameters and process flow of the complex lamination setup, contributing to a holistic understanding of the line's efficiency.
- **Drive System Configuration:** Hands-on experience with **Mitsubishi drive configurations**, enhancing my skills in power electronics and control systems.
- **Real-time Troubleshooting:** Actively participating in and resolving various technical challenges, refining my problem-solving and diagnostic abilities in a live industrial environment.

- **Comprehensive Machine Documentation:** Compiling and maintaining vital machine documentation, a cornerstone for operational continuity and long-term reliability.

Reflecting on Professional Development and Future Application

This internship has been instrumental in solidifying my practical foundation in **automation, electrical systems, and systematic troubleshooting logic**. Beyond technical proficiencies, I have also gained invaluable insights into the critical importance of **meticulous documentation, stringent safety protocols, and seamless inter-departmental coordination** within a dynamic industrial setting.

I extend my sincere gratitude for the exceptional support, guidance, and mentorship provided throughout this enriching experience. I am confident that the skills and perspectives gained here will be directly applicable and highly beneficial in my future endeavors. I look forward to the potential for our professional paths to intersect again.



Created By

Suyash Shukla

Verified By

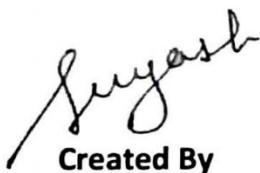
Saideep Kethireddi

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Created By

Suyash Shukla

Intern



Verified By
Gaideep Kethireddi
Senior Manager - HR

He has showed a great learning curve during this whole period. Curious, Enthusiastic and a person with good IQ. We wish him the best for his future!





REF: LCL: HRD: 25-26

June 20th, 2025TRAINING CERTIFICATE

This is to certify that **Mr. Suyash Shukla**, B.Sc. (Electronics and Computer Science) student of St Joseph's University, Bengaluru has completed his internship in our organization from 14th May 2025 to 20th June 2025.

We found him very hard-working and sincere during the training period.

We wish him every success in life.

For **LOHIA**CORP LTD.

Kuldeep Pathak
Executive Manager – HR

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